Bangladesh University of Professionals

Faculty of Science and Technology
Department of Information & Communication Technology (ICT)

Course Title: Database Management System Laboratory
Course Code: ICE 2204

Project Report

Project Name: Elderly Care & Emergency Alert System

Submitted to	**Submitted by**
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Introduction

The Elderly Care & Emergency Alert System is a database driven application designed to enhance the safety, health management, and well being of senior citizens. The system integrates emergency alert mechanisms, health tracking, and communication tools to ensure that elderly individuals receive timely medical attention and support during critical situations.

System User Overview

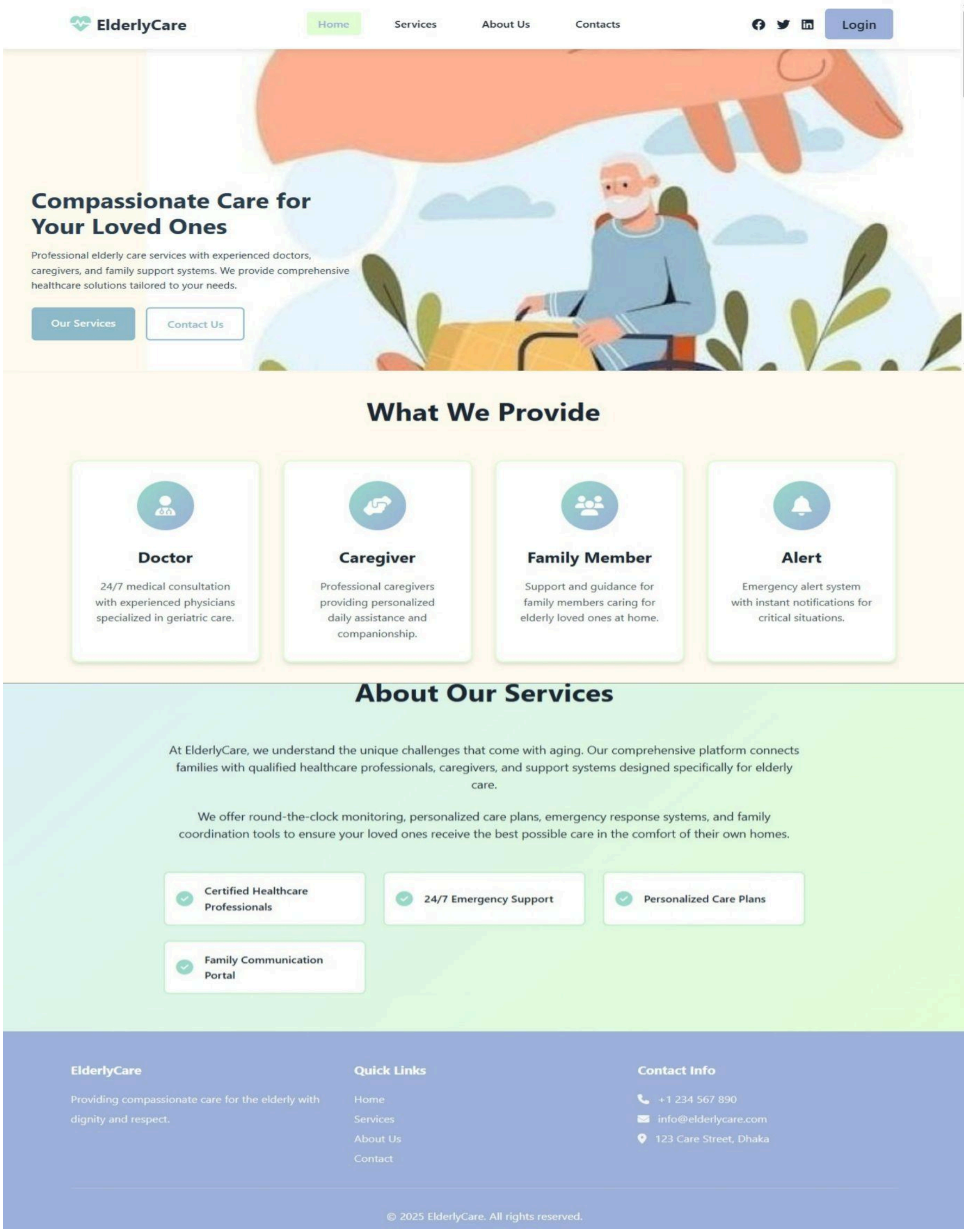
1. **Admin:**
   The Admin oversees the entire ElderlyCare system. They manage all users, elderly residents, caregivers, doctors, and family accounts. The admin monitors real time alerts, updates their status, and ensures that residents receive proper care. They maintain system data, review activity reports, and keep all records accurate and up to date. In short, the admin controls registrations, assignments, alerts, and overall system health.
2. **Elderly Person:**
   The main user of the system. They can view their health readings, upcoming appointments, medications, and care team details. They can schedule new appointments, review their records, and trigger emergency help when needed through the portal or a connected device.
3. **Family Member:**
   A trusted relative who actively monitors the elderly person's daily health, reviews vital updates, and stays informed through real time alerts. They can communicate with the care team, track appointments, and ensure timely responses during emergencies.
4. **Caregiver:**
   A trained helper who supports the elderly with daily care, monitors their health, and updates vital signs and tasks. The caregiver responds to alerts, ensures medications and appointments are managed, and communicates with doctors and family members to keep the patient safe and well.
5. **Doctor:**
   A medical professional who oversees the elderly patient's health by reviewing vital signs, analyzing alerts, and responding to emergencies. The doctor manages prescriptions, schedules appointments, and keeps treatment plans and medical records up to date to ensure proper ongoing care.

Functions & Operations

The primary functions along with their descriptions that is included in the project are depicted below :

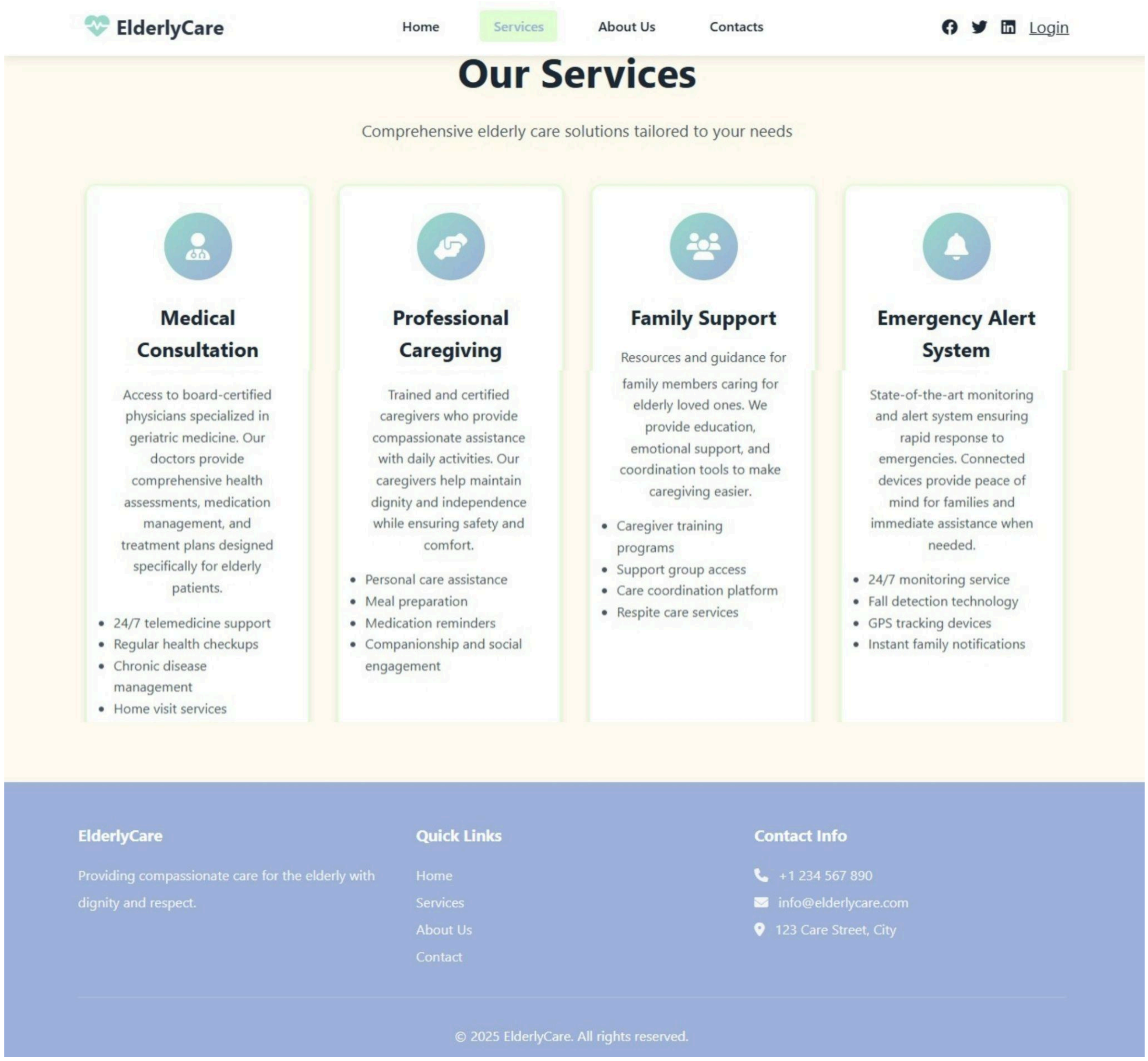
1.Index/Home Page

The homepage introduces ElderlyCare as a warm and supportive platform for seniors. It highlights compassionate care and connects users to doctors, caregivers, family support, and an emergency alert system. It also shows what the service provides and explains the mission of offering personalized elderly care with safety, communication, and 24/7 support. The page ends with quick links and contact information.



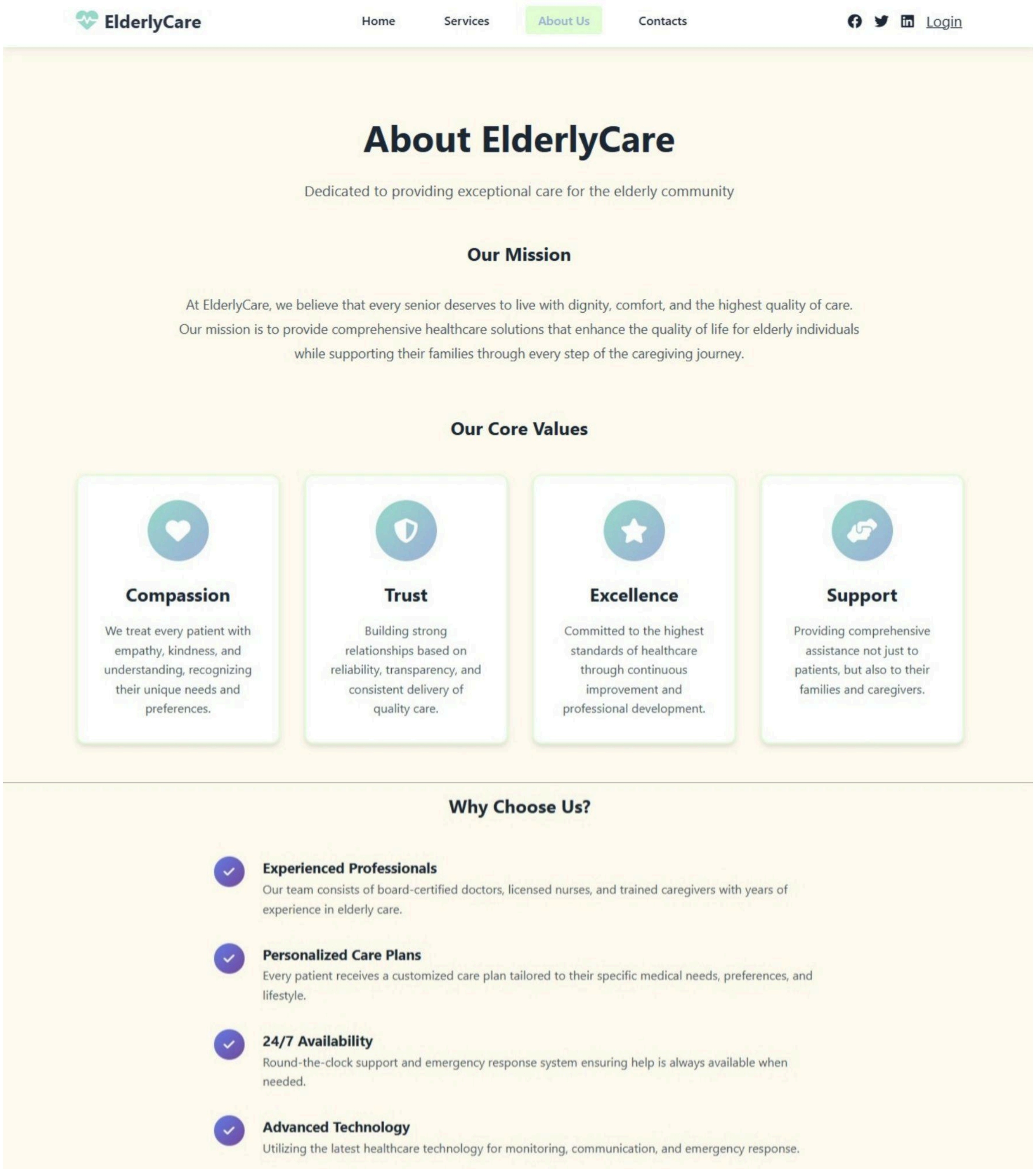
2.Servies

The services page explains the main care solutions ElderlyCare offers. It includes medical consultations from certified doctors, professional caregiving for daily tasks, family support programs, and a modern emergency alert system with monitoring and instant notifications. Each service card gives a brief, easy to understand overview of what families and seniors can expect.



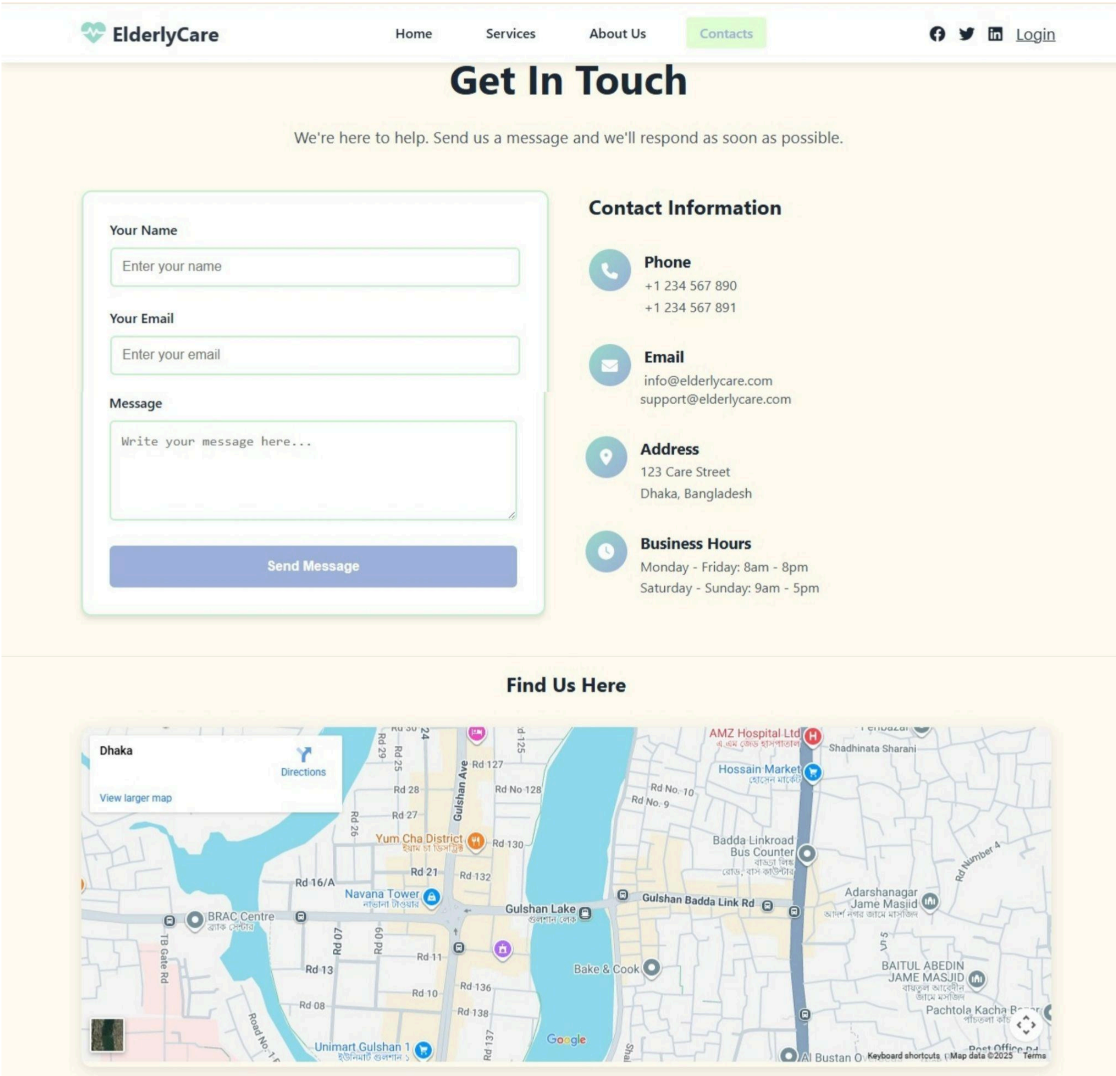
3.About Us

The About Us page introduces ElderlyCare as a service dedicated to improving the lives of seniors. It explains the mission of offering dignified, comfortable, and high quality care while supporting families. It highlights core values—compassion, trust, excellence, and support—using simple, easy to read sections. The page also shares reasons to choose ElderlyCare, such as experienced professionals, personalized care plans, 24/7 availability, and modern healthcare technology.



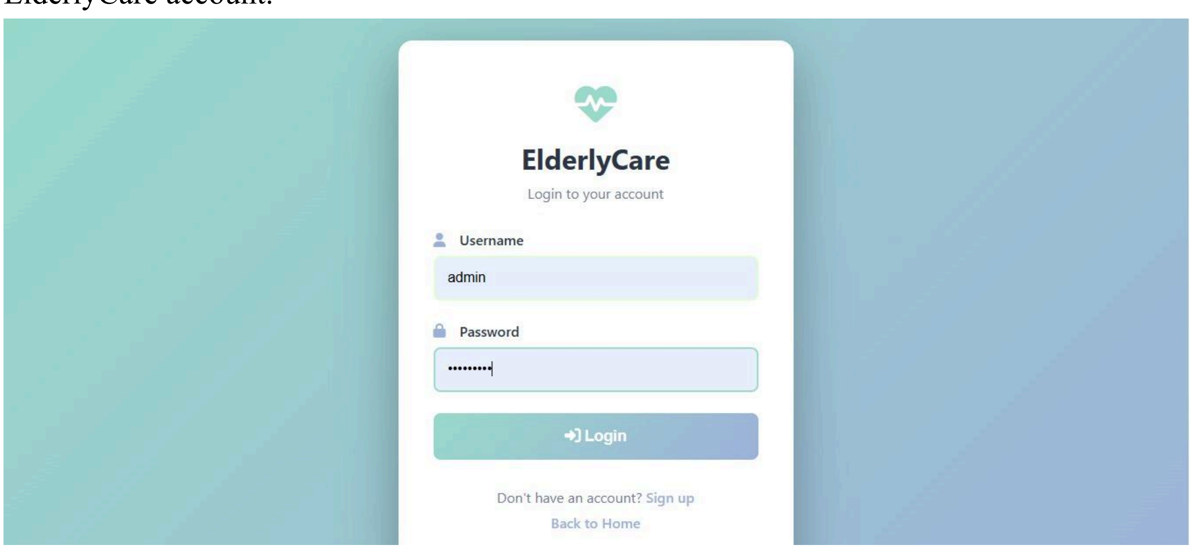
4.Contact

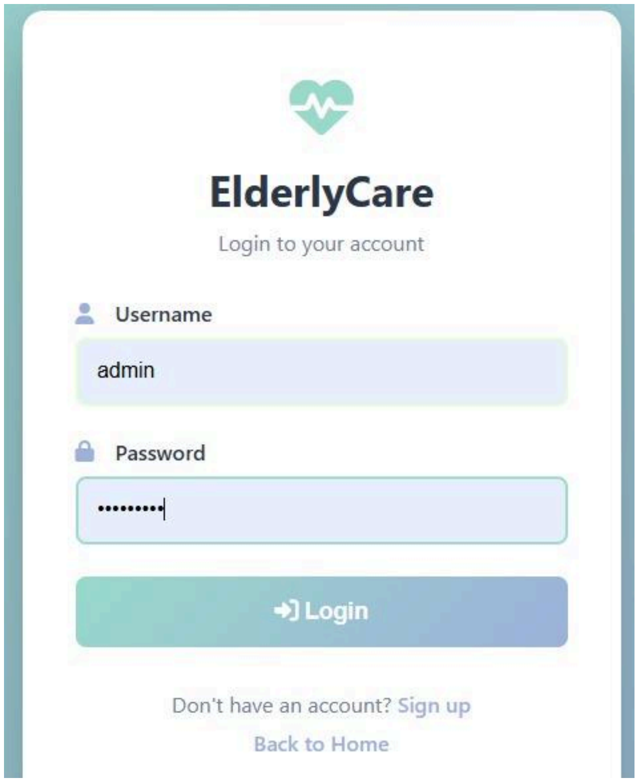
The Contacts page makes it easy for anyone to get in touch. It includes a clean contact form for name, email, and message, along with clear contact details like phone numbers, email addresses, location, and business hours. A map at the bottom shows exactly where the center is located, helping visitors find it quickly.



5. User Login Page

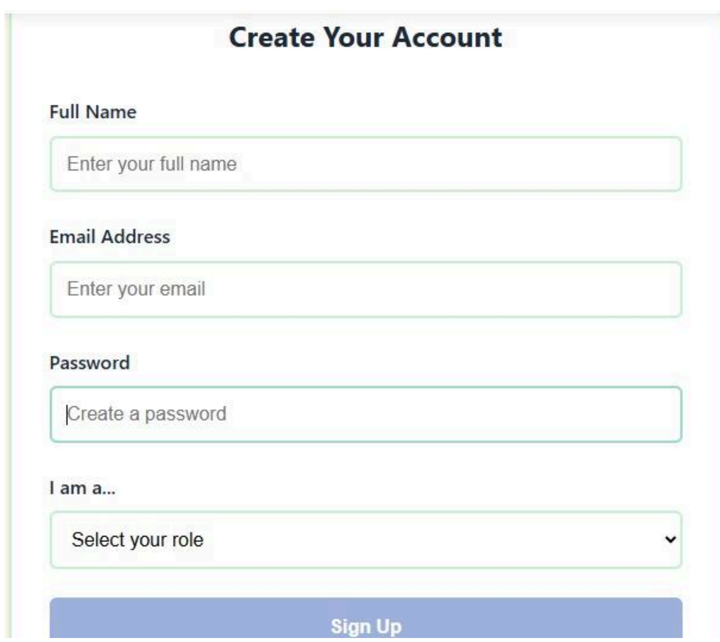
A professionally designed login interface that includes fields for entering a username and password, a secure login button, and navigation links for account registration and returning to the homepage. The page features a clean layout that ensures easy and efficient user access to their ElderlyCare account.

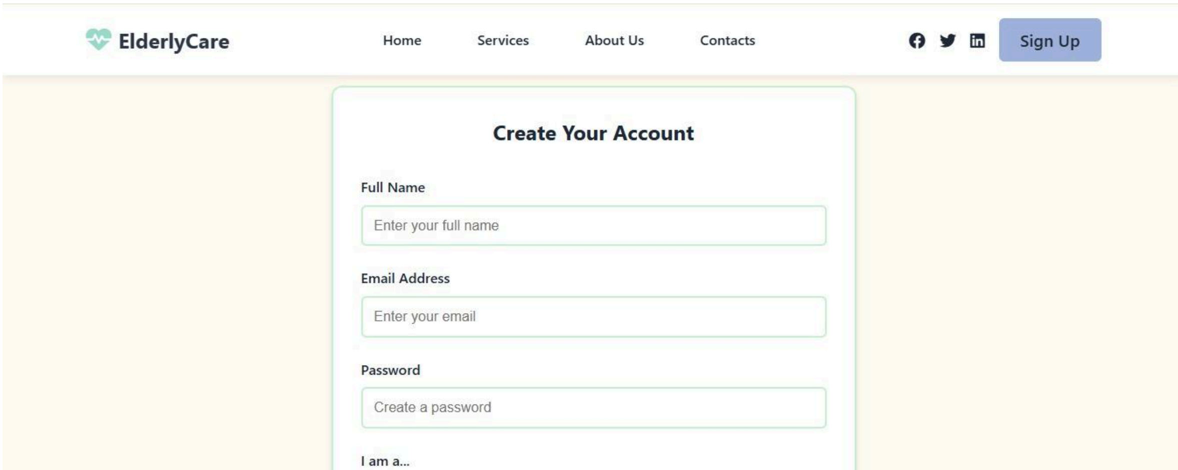


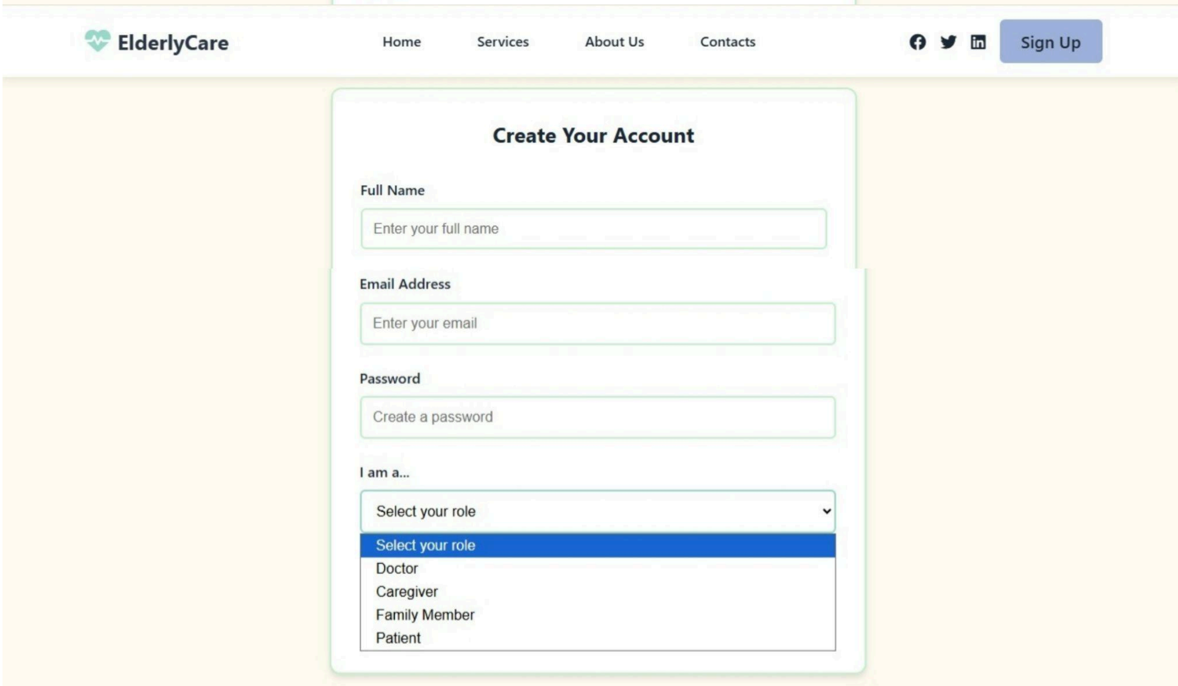


6.Sign Up Page

A structured account creation page containing input fields for full name, email address, and password, along with a role selection dropdown offering options such as Doctor, Caregiver, Family Member, and Patient. The page includes a clear sign up button and a navigation menu, providing a formal and organized onboarding experience for new ElderlyCare users.



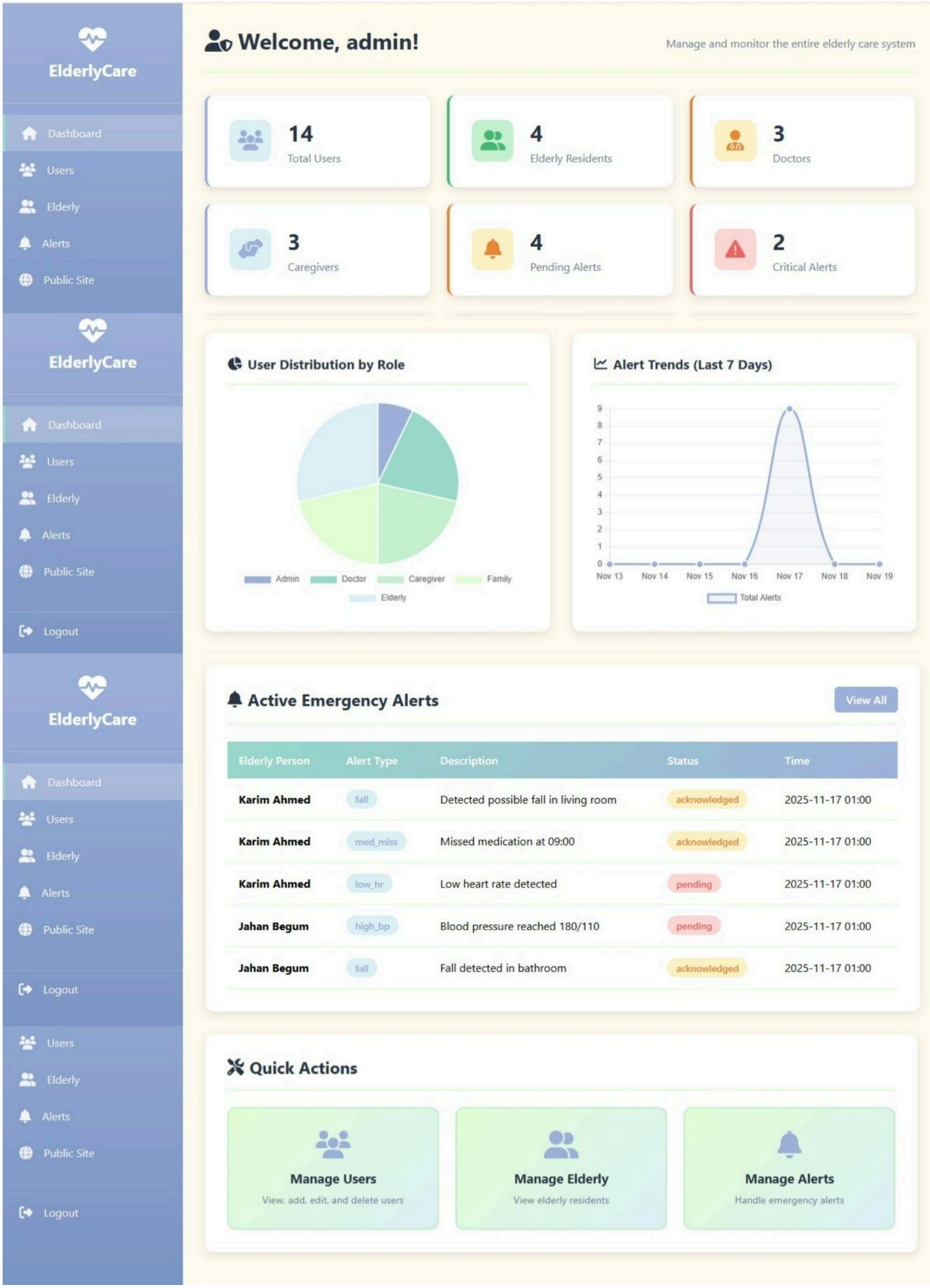




7.Admin Dashboard

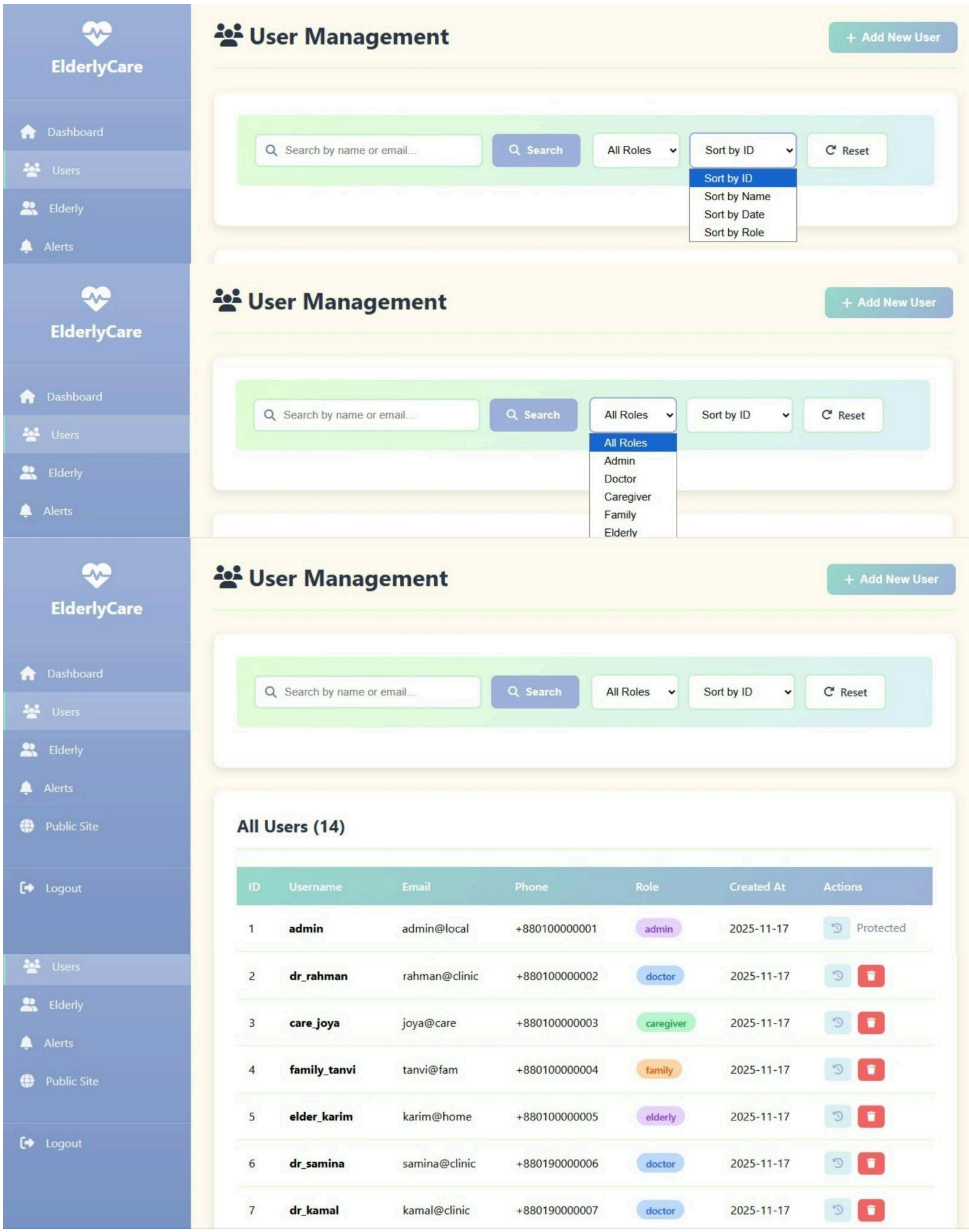
Dashboard Overview Page

The admin dashboard provides a clear summary of the entire ElderlyCare system. It shows quick statistics such as total users, elderly residents, doctors, caregivers, pending alerts, and critical alerts. The dashboard includes a pie chart showing user distribution by role and a line graph showing alert trends over the last 7 days. A section of active emergency alerts displays real time issues like falls, missed medication, low heart rate, or high blood pressure, along with their status (acknowledged or pending). At the bottom, quick action buttons let the admin manage users, elderly residents, and alerts efficiently.


ElderlyCare
Dashboard
Users
Elderly
Alerts
Public Site
Logout
Welcome, admin!
Manage and monitor the entire elderly care system
14
Total Users
4
Elderly Residents
3
Doctors
3
Caregivers
4
Pending Alerts
2
Critical Alerts
User Distribution by Role
Admin
Doctor
Caregiver
Family
Elderly
Alert Trends (Last 7 Days)
Nov 13
Nov 14
Nov 15
Nov 16
Nov 17
Nov 18
Nov 19
Total Alerts
Active Emergency Alerts
View All
Elderly Person
Alert Type
Description
Status
Time
Karim Ahmed
fall
Detected possible fall in living room
acknowledged
2025-11-17 01:00
Karim Ahmed
med_miss
Missed medication at 09:00
acknowledged
2025-11-17 01:00
Karim Ahmed
low_hr
Low heart rate detected
pending
2025-11-17 01:00
Jahan Begum
high_bp
Blood pressure reached 180/110
pending
2025-11-17 01:00
Jahan Begum
fall
Fall detected in bathroom
acknowledged
2025-11-17 01:00
Quick Actions
Manage Users
View, add, edit, and delete users
Manage Elderly
View elderly residents
Manage Alerts
Handle emergency alerts

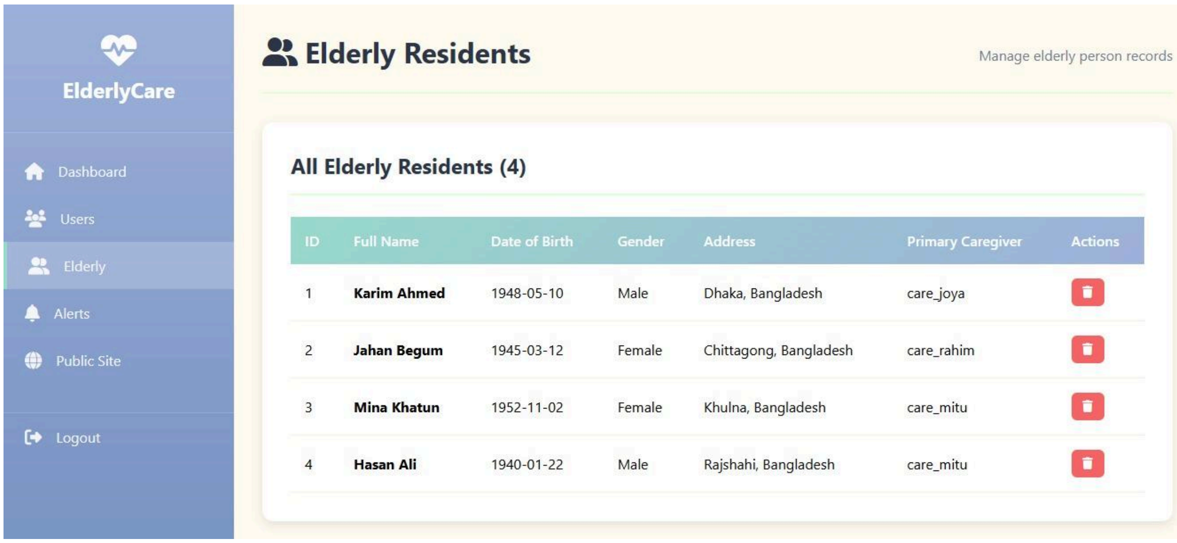
User Management Page

The User Management page allows the admin to view and manage all users in the system. It includes a search bar, role filter, and sorting options (ID, Name, Date, Role) to quickly find specific users. The page lists all users with details such as username, email, phone number, role, and account creation date. Each row includes action buttons for editing or deleting users. A button at the top lets the admin add new users easily. Overall, this page helps the admin monitor and control every user in the ElderlyCare system.



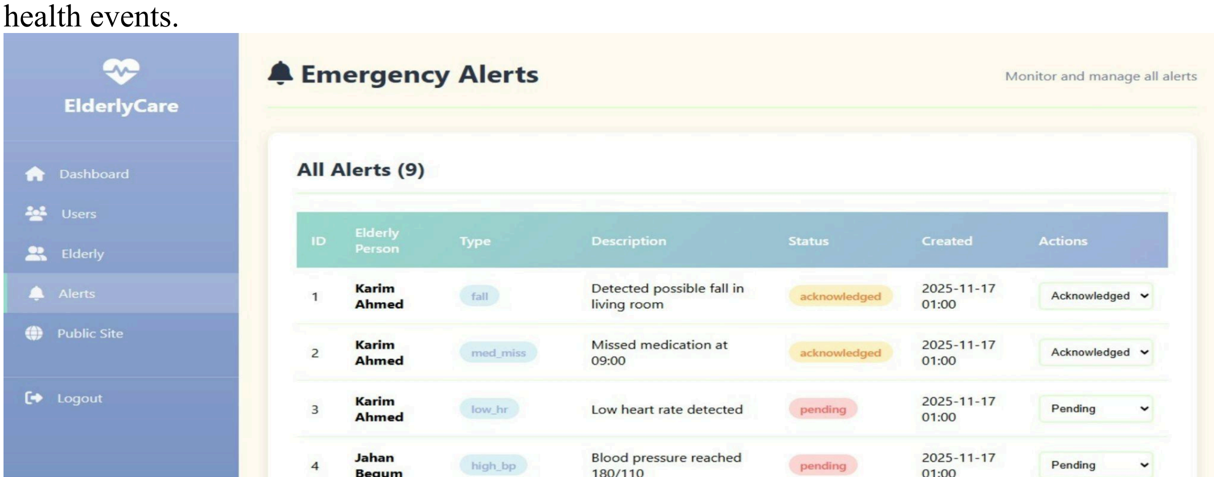
Elderly Residents Page

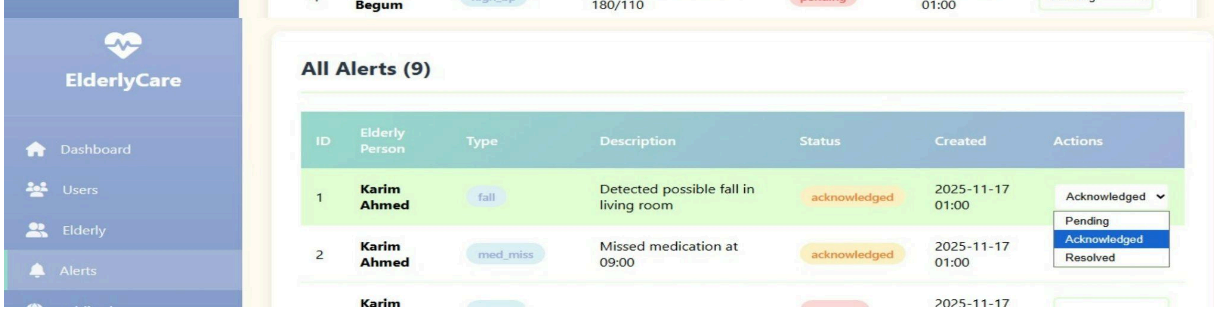
The Elderly Residents page allows the admin to view and manage all elderly individuals registered in the system. It displays each resident's full name, date of birth, gender, address, and assigned primary caregiver. A delete action button appears for managing records when needed. This page helps the admin keep track of all residents and their caregiving assignments in one place.



Emergency Alerts Page

The Emergency Alerts section shows all alerts generated by elderly residents, including fall detection, missed medication, low heart rate, and high blood pressure. Each alert entry includes the elderly person's name, alert type, description, status (acknowledged or pending), and the date/time it was created. The admin can update the alert status using a dropdown (Acknowledged, Pending, Resolved). This page helps monitor and respond quickly to urgent health events.





8.Elderly Dashboard

Dashboard Overview Page

1. Welcome Header & Elderly Profile Card

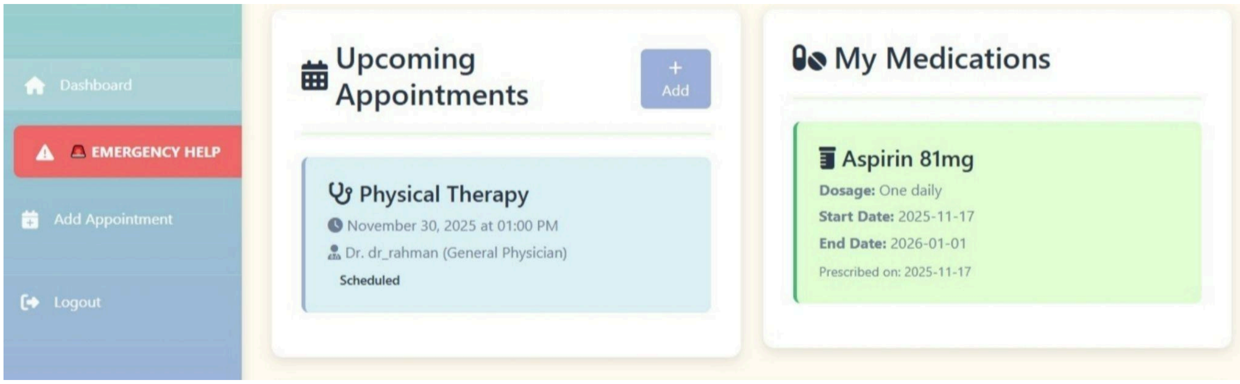
A friendly greeting that shows the patient's name and today's date, helping the user feel personally acknowledged when they log in.Displays essential personal details such as: Name and age,Emergency contact,Location.This gives caregivers and the patient a quick snapshot of identity and emergency information.

2. Current Health Status

A set of easy-to-read health indicators: **Heart Rate,Blood Pressure,Blood Sugar,Body Temperature.**Each metric is clearly displayed with icons, helping users quickly check their vital signs.

3. Upcoming Appointments & My Medications

Shows the next scheduled medical visit, including: Appointment type (e.g., Physical Therapy),Date and time,Assigned doctor.There's also an **Add** button for making new appointments easily. Lists active prescriptions such as: Medication name (e.g., Aspirin 81mg),Dosage instructions,Start and end date.This helps patients manage their daily medication routines.

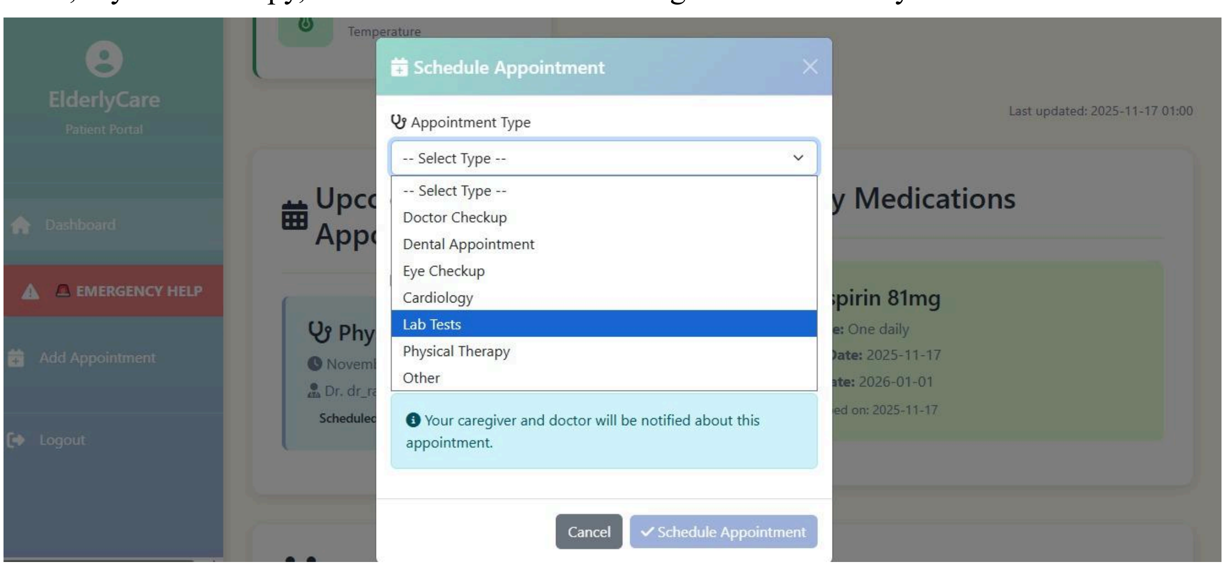


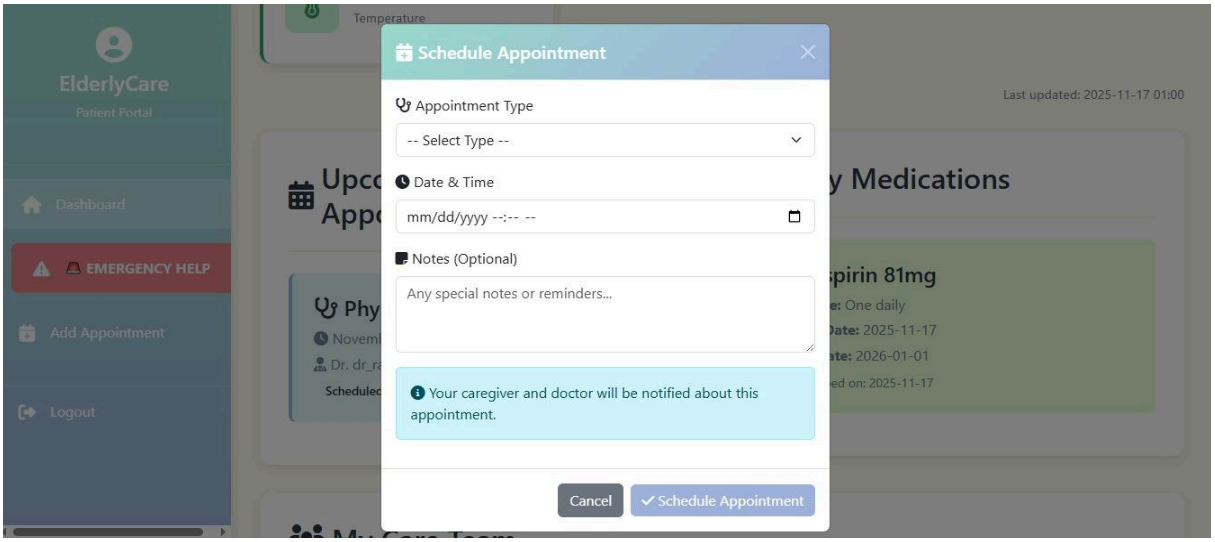
4. My Care Team

Introduces the patient's support network, including: **Primary caregiver:** name, phone, email, shift,**Doctor:** name, specialty, phone, email.This section ensures the patient and family know who is responsible for care.

5. Emergency Help Sidebar,Appointment Scheduling Popup & Appointment Type Dropdown

A clearly highlighted button for immediate assistance, ensuring quick access in urgent situations.A clean form that lets the patient select: Appointment type,Date and time,Optional notes.It also notes that both caregiver and doctor will be notified automatically.A simple menu that offers choices such as: Doctor Checkup,Dental Appointment,Eye Checkup,Cardiology,Lab Tests,Physical Therapy,Other. This makes scheduling flexible and easy.



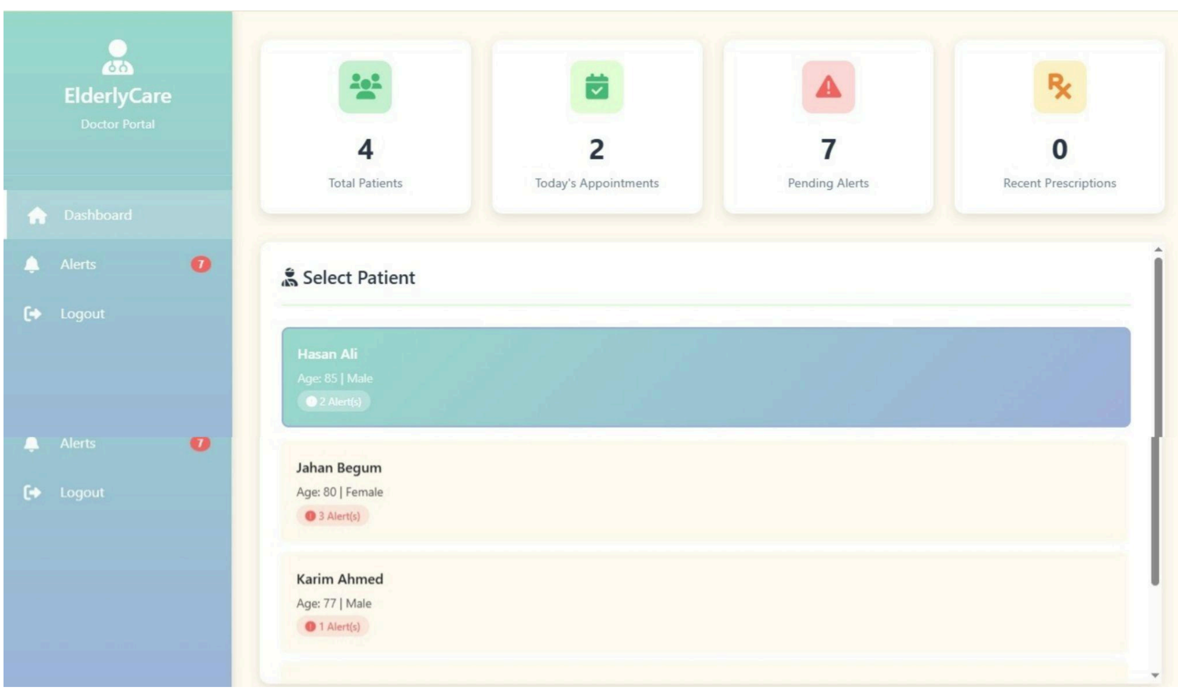


9.Doctor Dashboard

Dashboard Overview Page

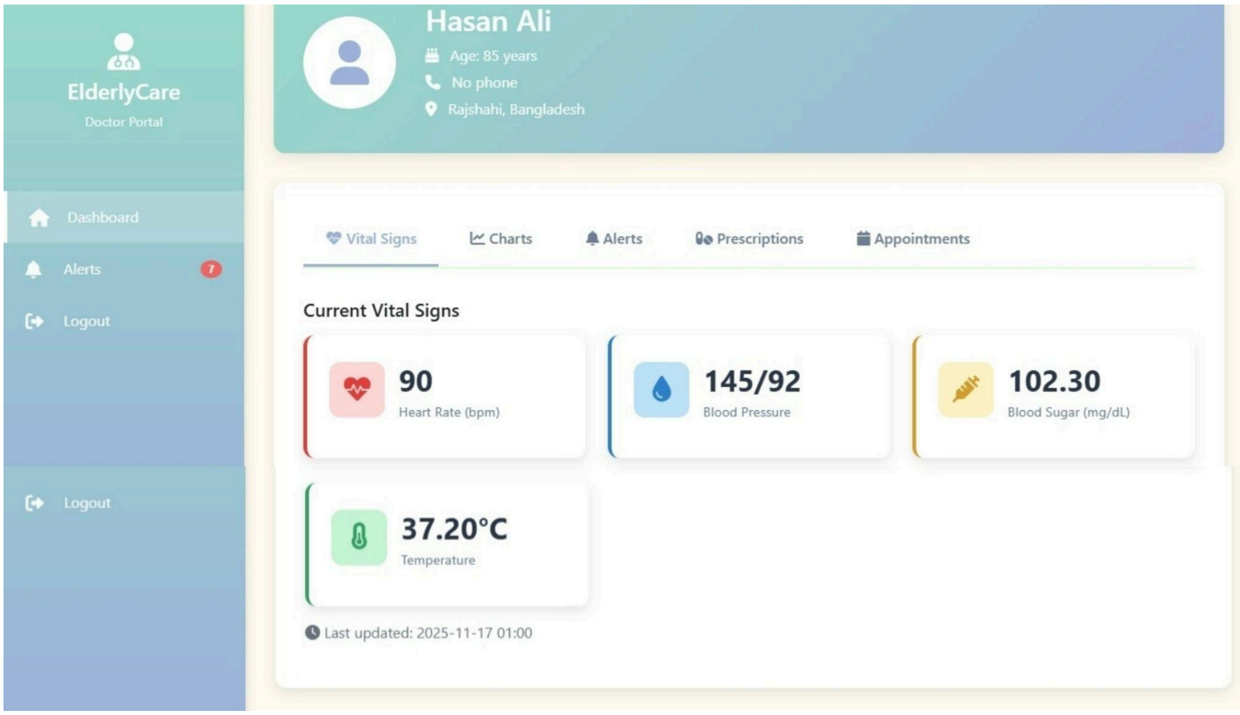
1. Dashboard Summary Cards & Select Patient List

A quick overview showing:- **Total Patients:** How many patients the doctor is currently managing;**Today's Appointments:** Number of appointments scheduled for the day;**Pending Alerts:** Health alerts that need immediate attention and **Recent Prescriptions:** Medications recently issued to patients. This gives the doctor a fast snapshot of workload and urgent tasks.A list of all registered patients, each card showing: Name, age, and gender; Number of active health alerts and Highlighted cards (e.g., the selected patient) for easy navigation.This section helps doctors quickly choose a patient to review their details or respond to alerts.



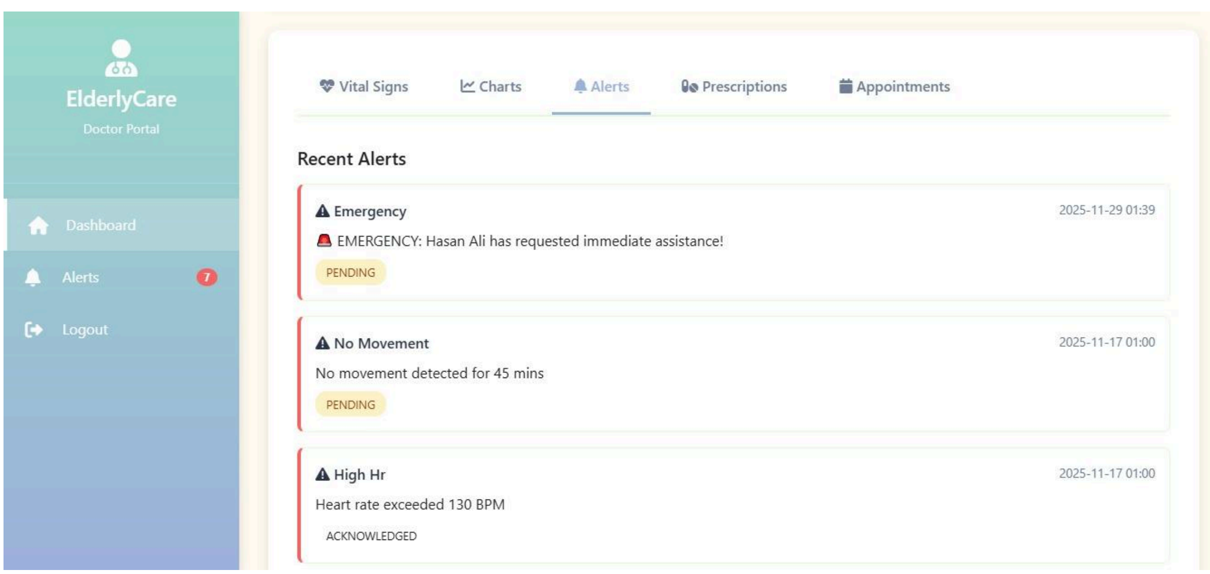
2.Patient Profile & Vital Signs

Shows the patient’s basic information—name, age, contact status, and location—so the doctor immediately knows who they are reviewing.Displays the patient’s latest health readings, including **heart rate**, **blood pressure**, **blood sugar**, and **temperature**. This gives the doctor a quick snapshot of the patient’s current condition.



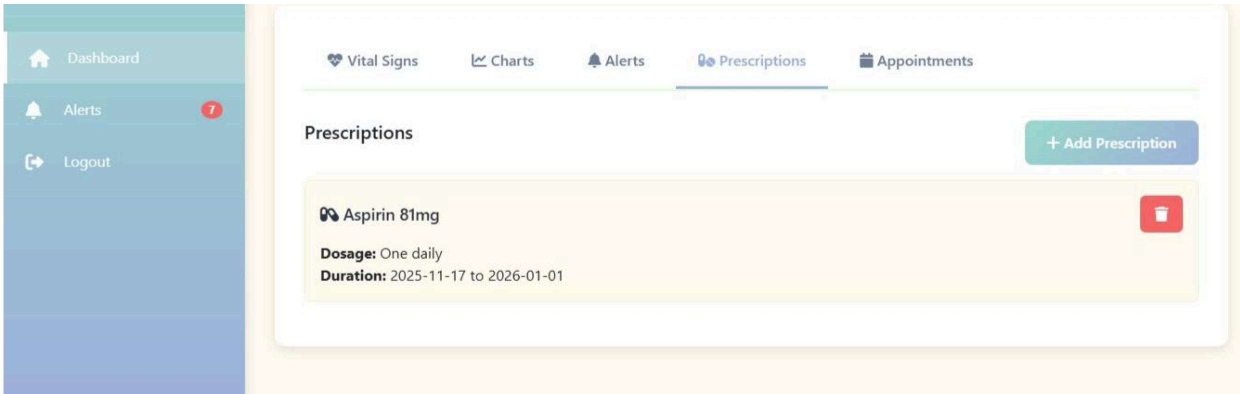
Recent Alerts

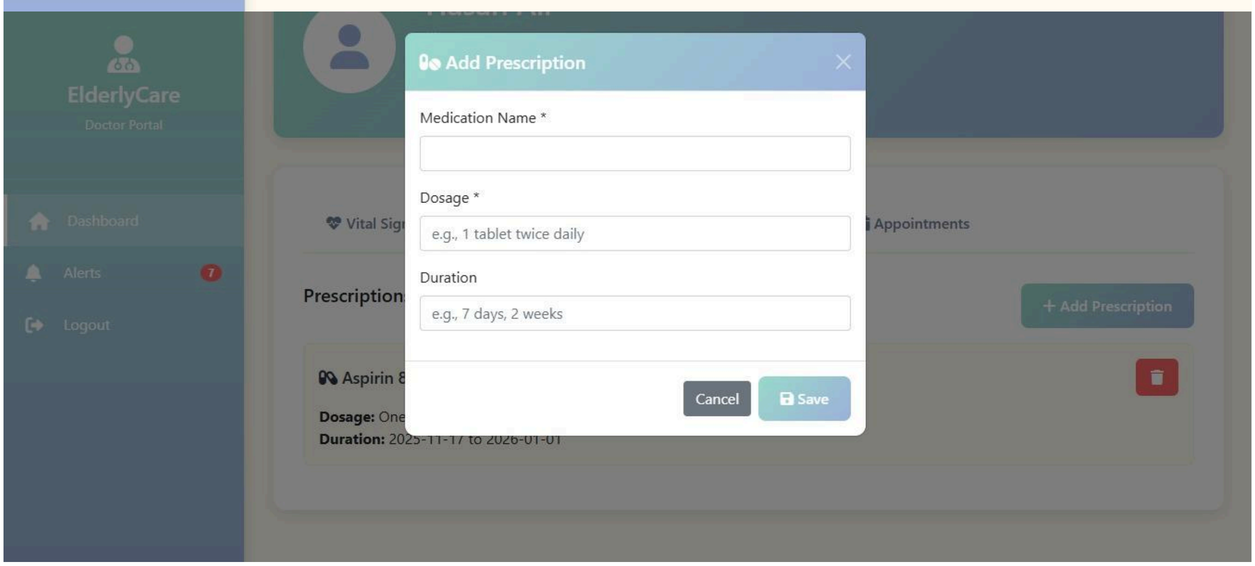
Lists the most recent health alerts raised for the patient, such as **emergencies**, **no-movement warnings**, or **abnormal heart rate**. Each alert includes a time and status so the doctor can see what needs attention.



Prescriptions & Add Prescription

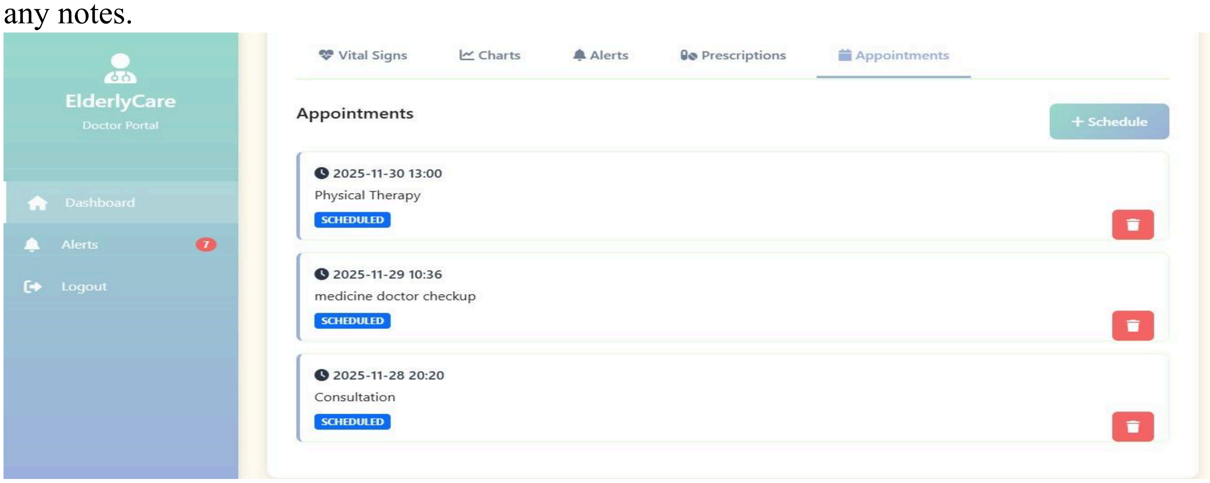
Shows all active medications prescribed to the patient, including **dosage** and **duration**. Doctors can add new prescriptions or remove outdated ones through the modal form.A quick form where the doctor enters medication name, dosage, and treatment duration to issue a new prescription.

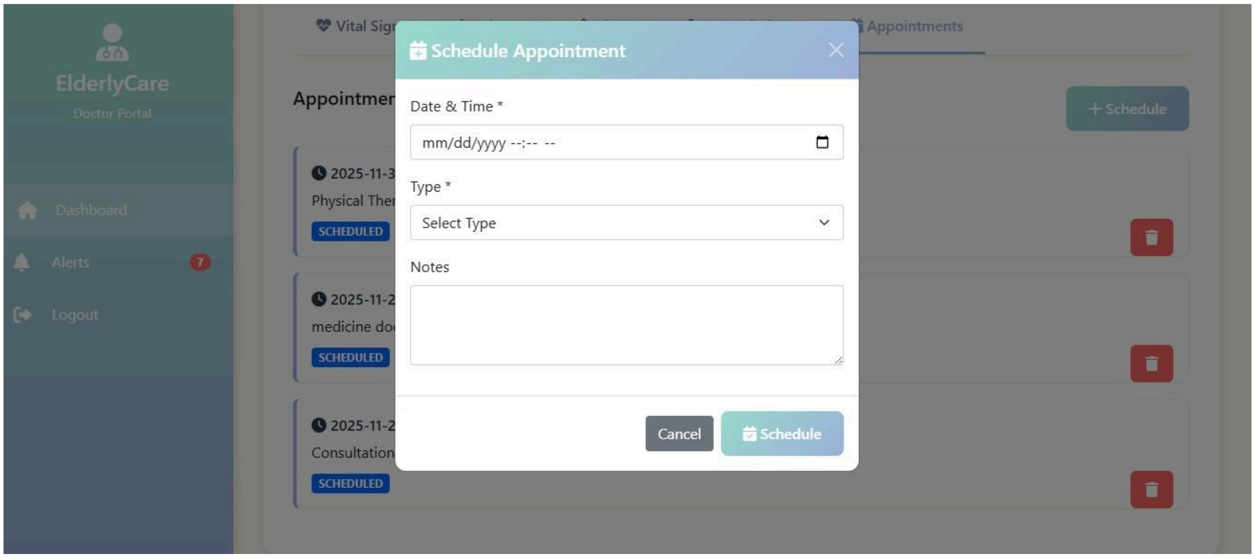




Appointments & Schedule Appointment

Displays all upcoming scheduled appointments, including **date**, **time**, and **visit type**. Doctors can schedule new appointments or delete existing ones using the appointment form.A simple form used to book a new appointment, allowing the doctor to set the date, time, appointment type, and any notes.

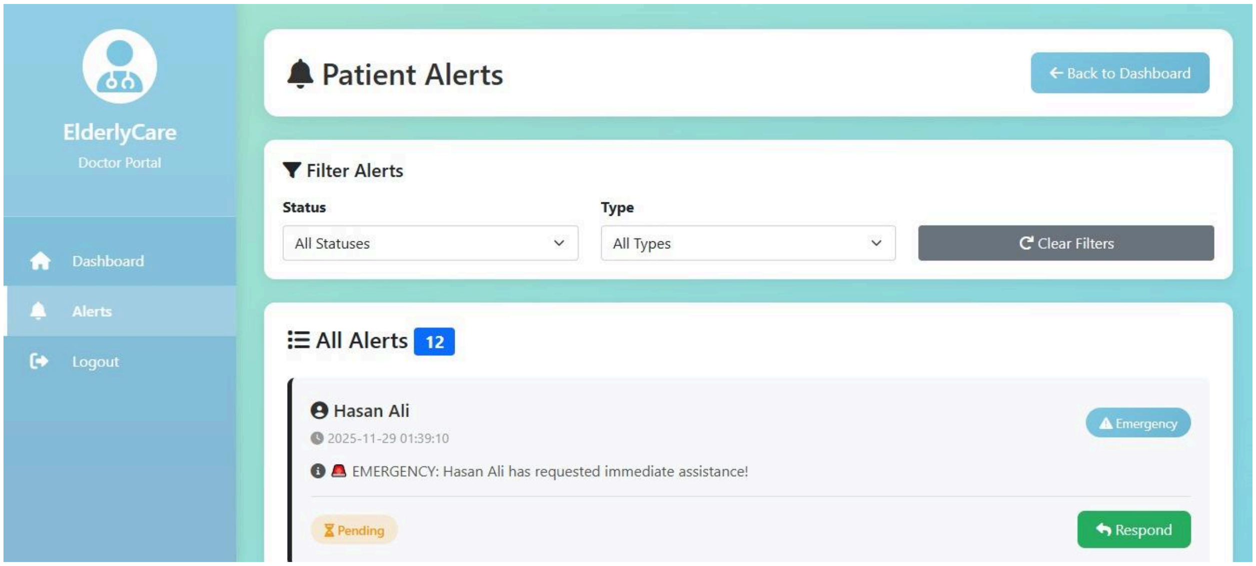




Alert Page

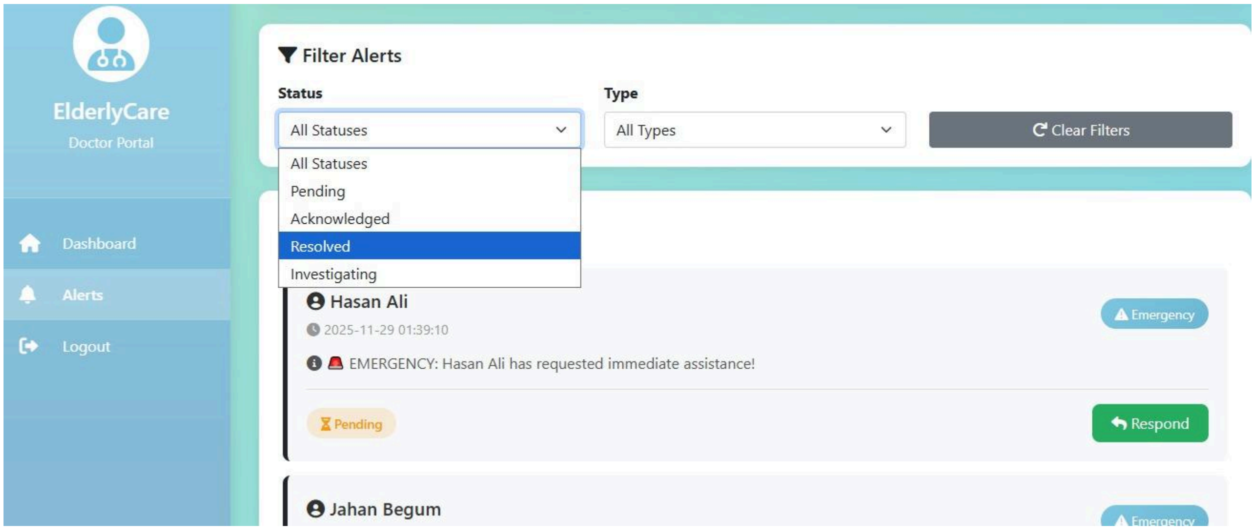
Patient Alerts

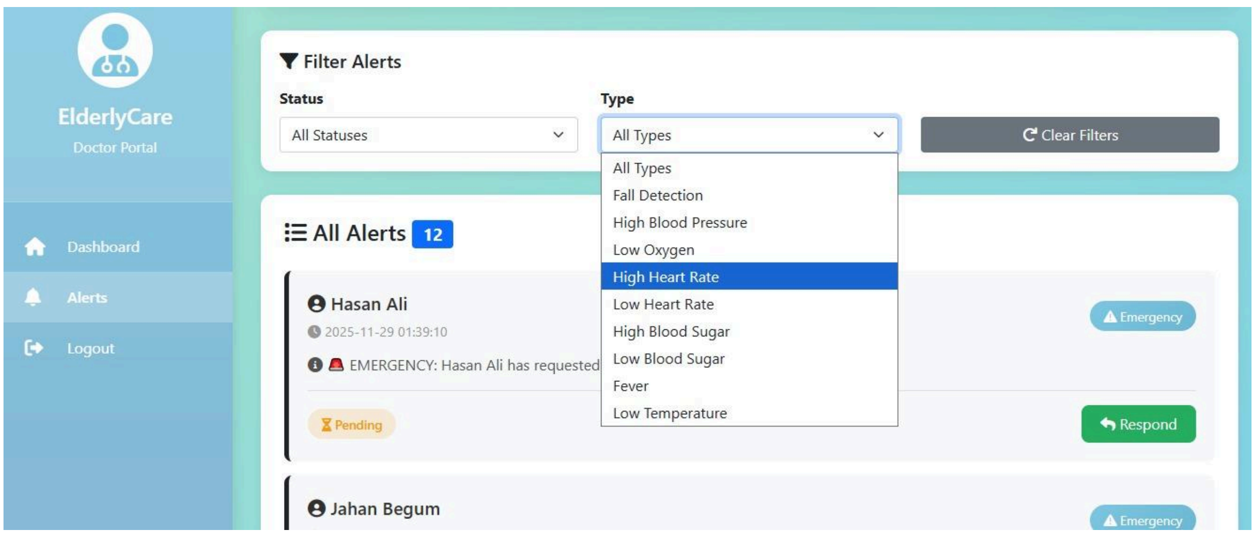
A central page where the doctor can quickly review all emergency and health-related alerts triggered by elderly patients.



Filter Alerts

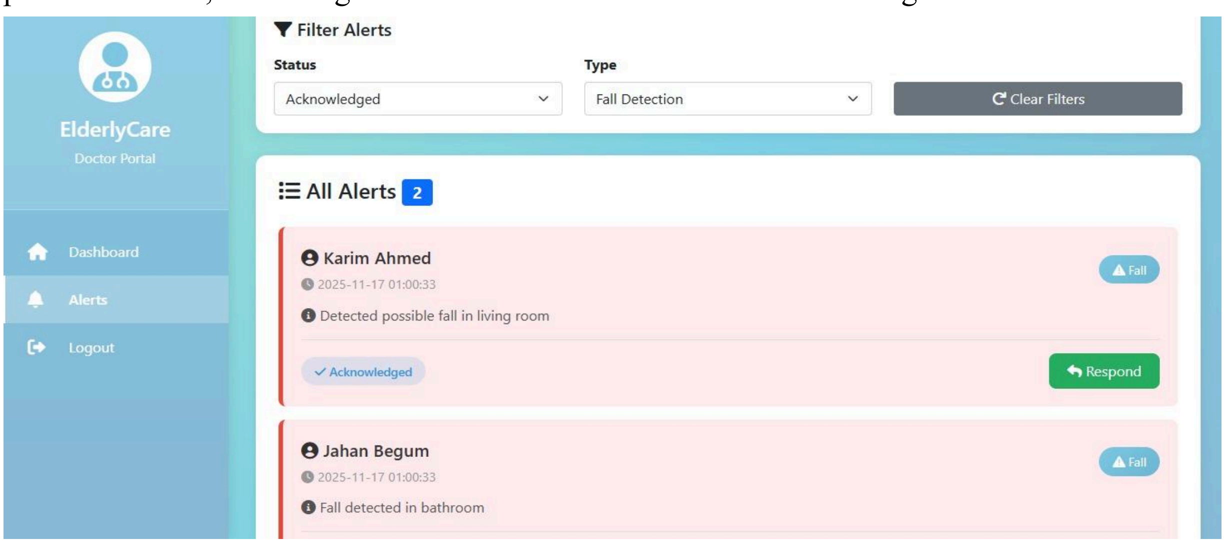
A simple filter panel that lets the doctor narrow alerts by **status** (e.g., Pending, Acknowledged, Resolved, Investigating) and **alert type** (e.g., Fall Detection, High Blood Pressure, Low Oxygen, Fever). This helps the doctor quickly focus on the most urgent cases.





All Alerts

A list of all patient alerts, showing: **Patient name, Timestamp, Alert type** (e.g., Emergency, High Heart Rate), **Current status,** A **Respond** button to take action immediately.This section provides a clear, chronological overview of all health incidents needing medical attention.

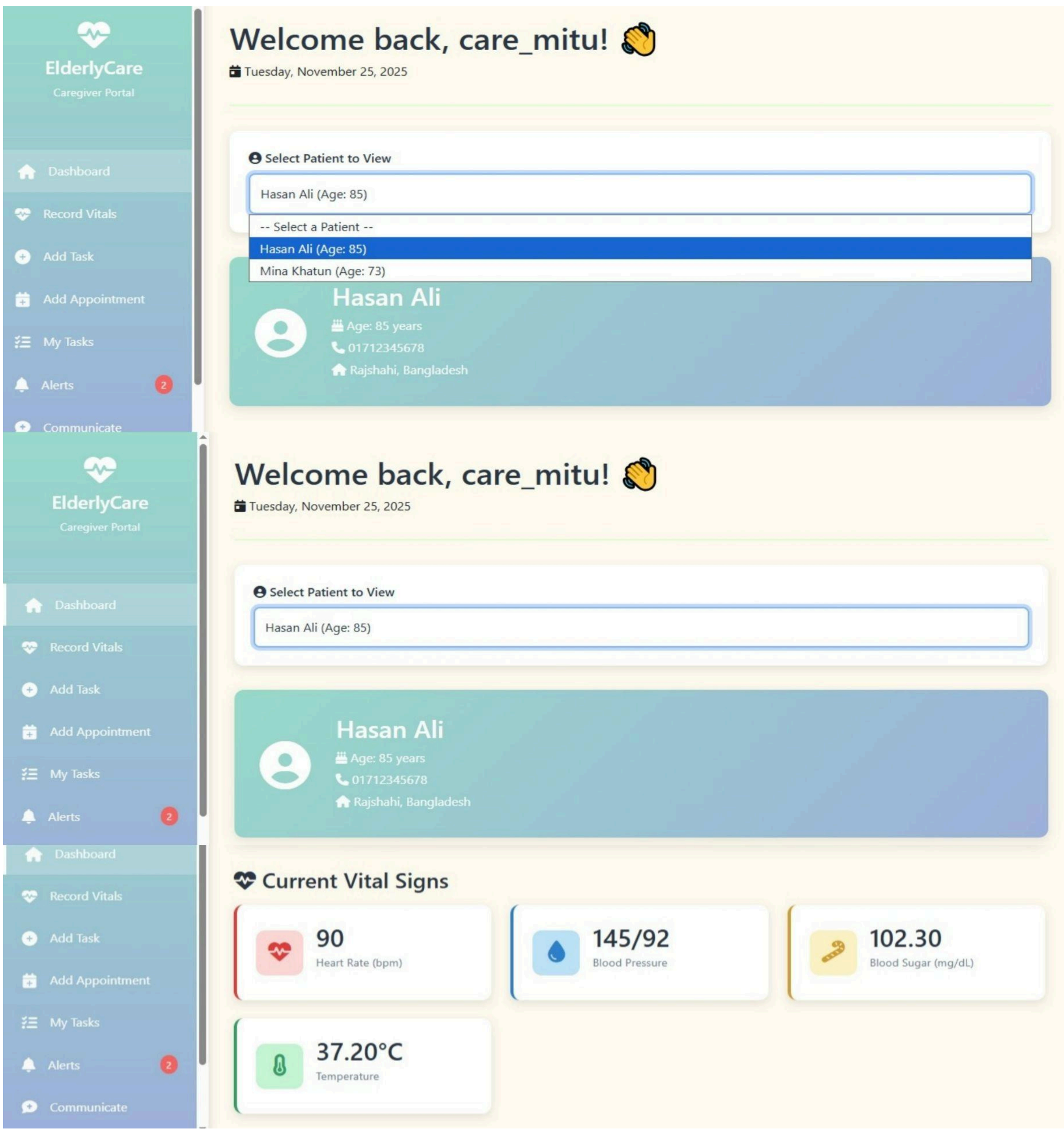


10.Caregiver Dashboard

Dashboard Overview Page

1. Welcome Header,Select Patient to View & Patient Overview Card

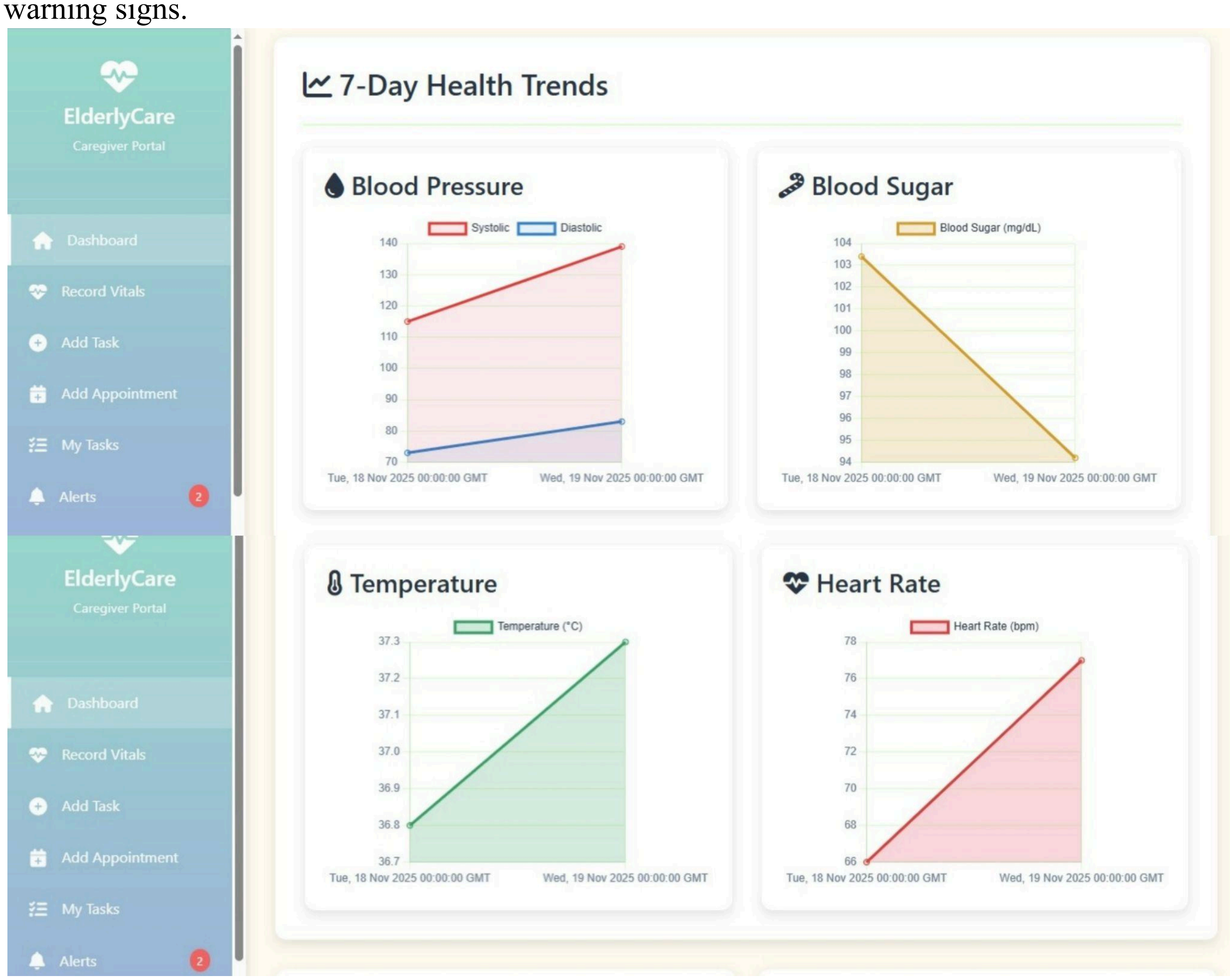
A friendly greeting that shows the caregiver’s name and today’s date, helping them quickly settle into the dashboard.A dropdown menu where the caregiver chooses which elderly person’s data to view. Once selected, all vitals, tasks, alerts, and history update instantly for that patient.Displays the patient’s name, age, photo placeholder, and basic health info at a glance—helping caregivers identify the right profile quickly.



Health Monitoring Sections

2. Current Vital Signs & 7-Day Health Trends

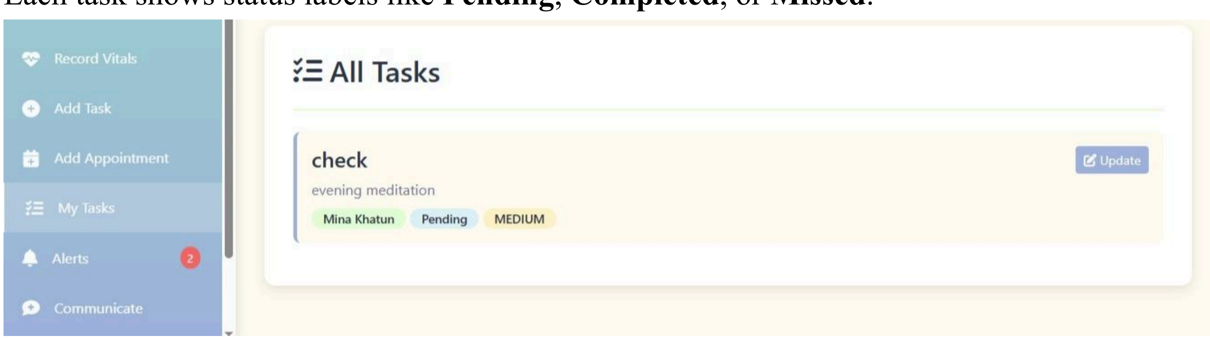
A real-time snapshot of the patient's latest health readings: Heart Rate,Blood Pressure,Blood Sugar,Temperature.Simple, color-coded boxes make it easy to spot anything unusual.Clear mini-graphs showing how key vitals changed through the week: Blood Pressure,Blood Sugar,Temperature,Heart Rate.This helps caregivers understand patterns and detect early warning signs.



Task & Appointment Management

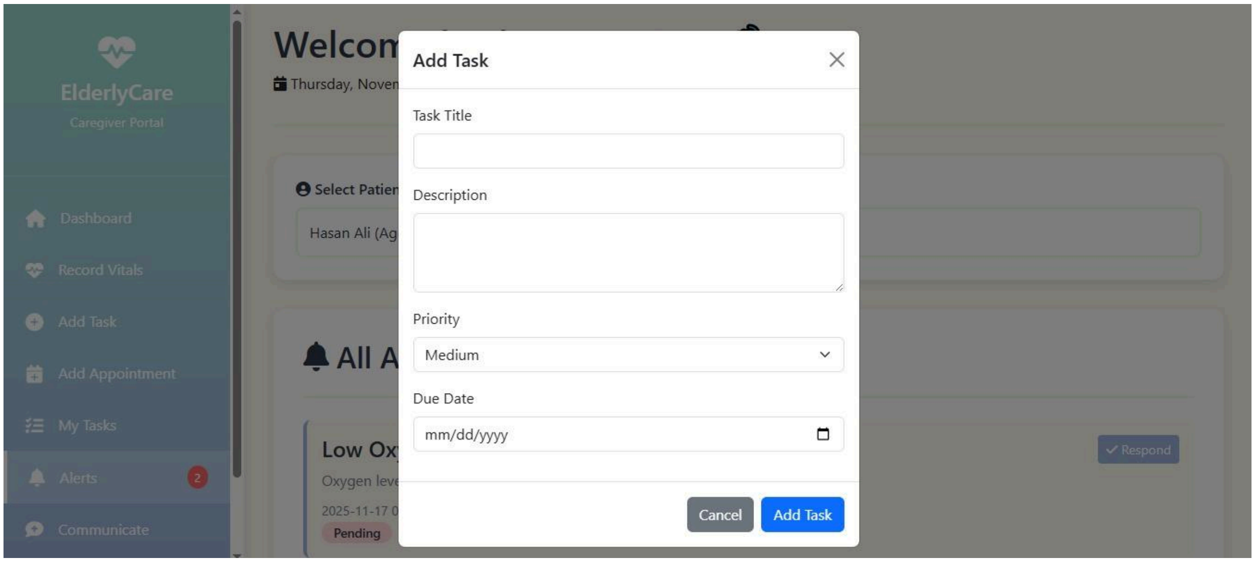
3. All Tasks

A list of all duties assigned for the patient such as medication, checkups, reminders, and more. Each task shows status labels like **Pending**, **Completed**, or **Missed**.



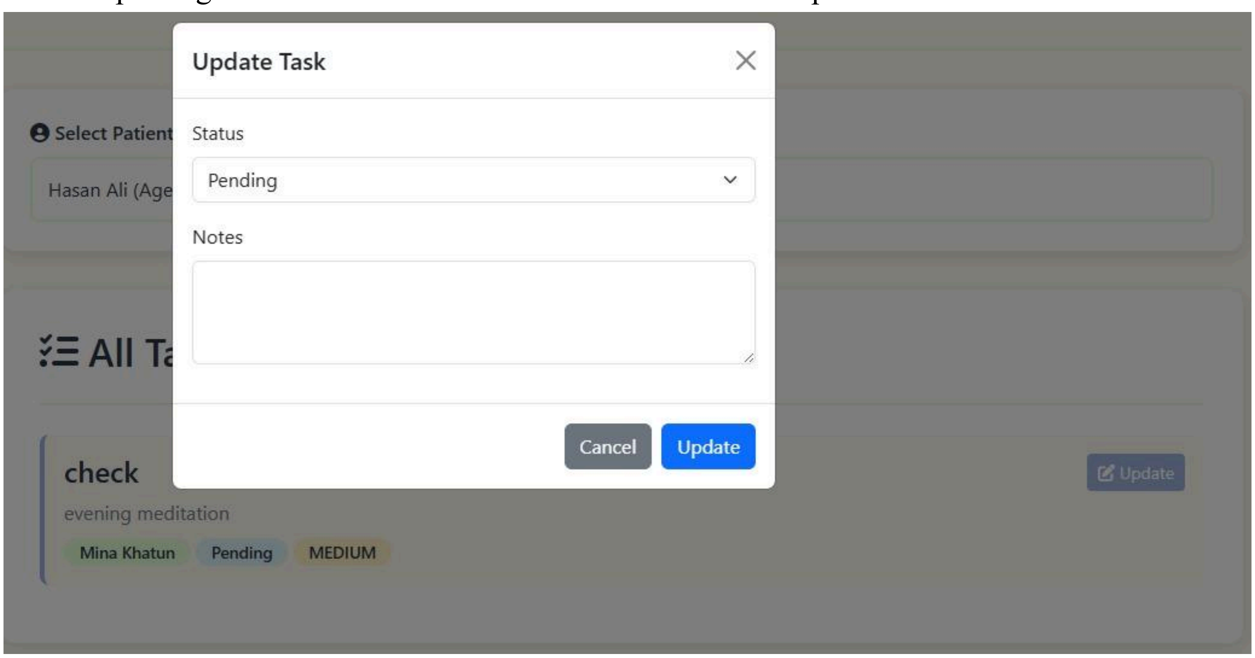
4. Add Task

A simple form to add new caregiving tasks with title, description, priority level, and due date. Helps ensure nothing important is overlooked.



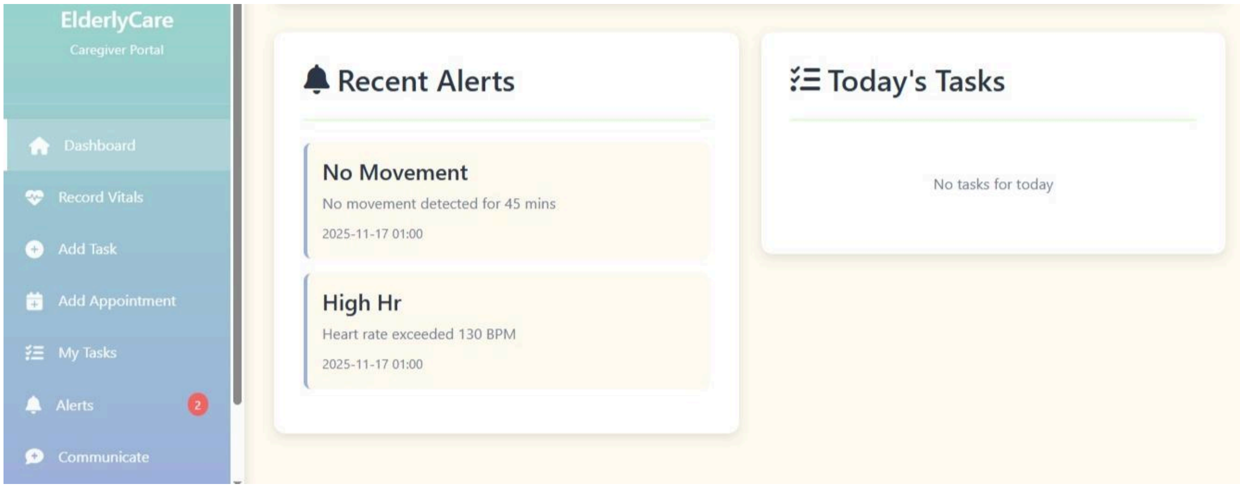
5. Update Task

Allows updating task status and notes as care activities are completed or reviewed.



6. Today's Tasks

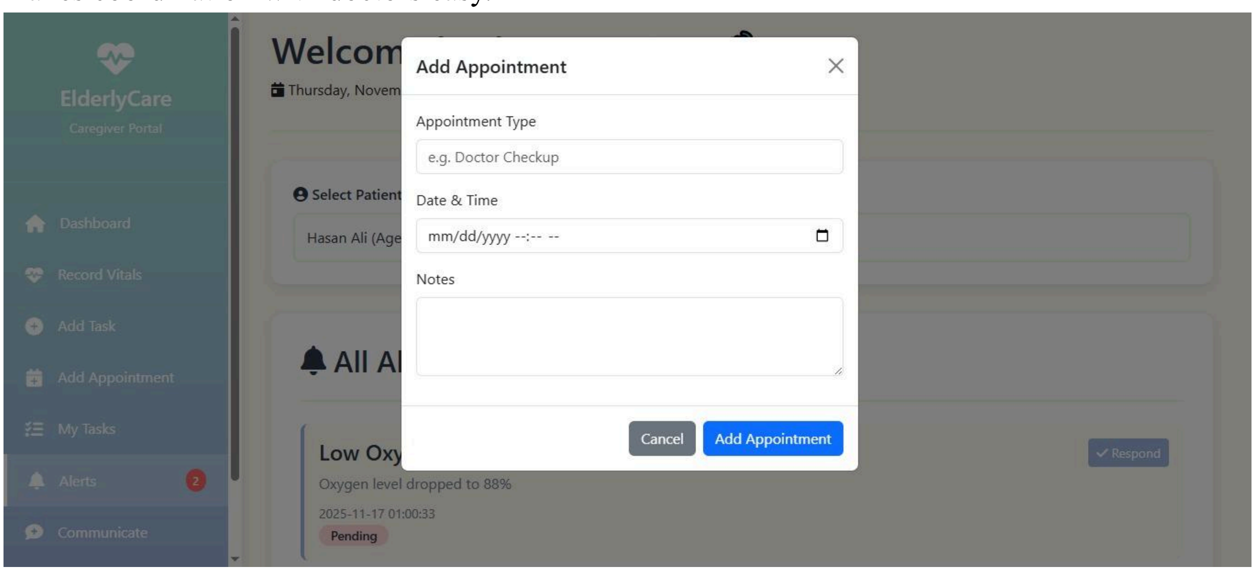
A compact panel showing tasks due *today*, keeping caregivers focused on same-day responsibilities.



Appointment Management

7. Add Appointment

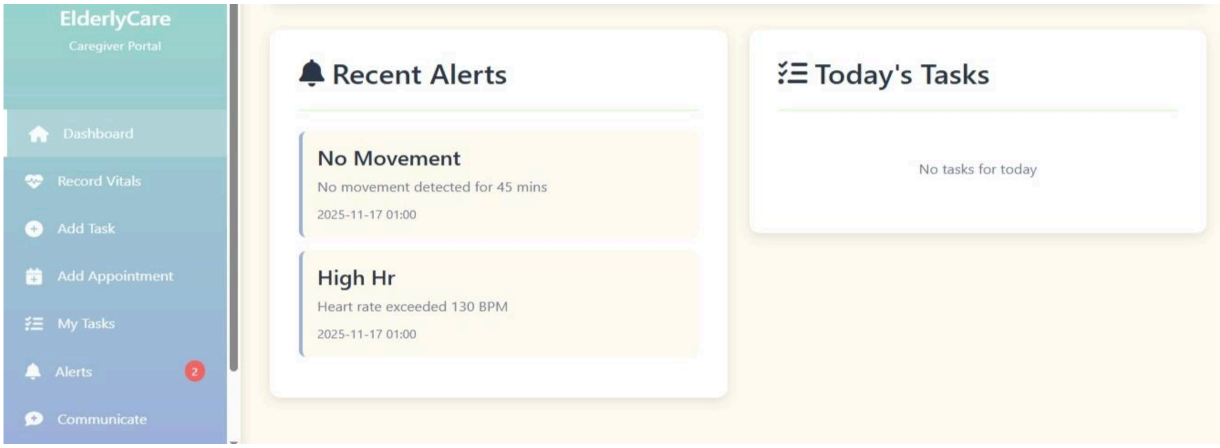
A clean form for scheduling medical appointments, selecting type, date, time, and optional notes. Makes coordination with doctors easy.



Safety & Alert

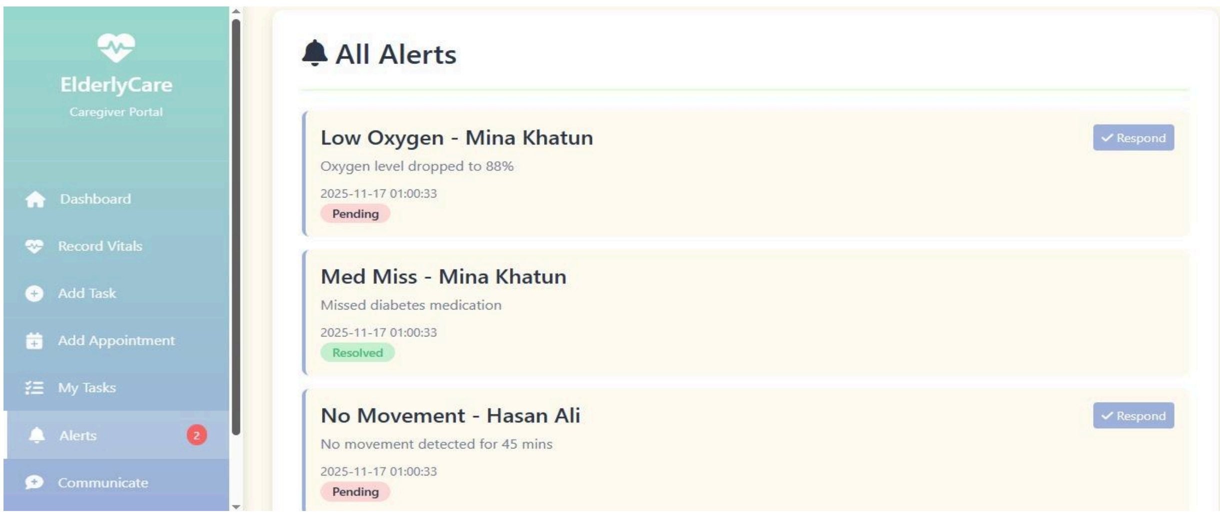
8. Recent Alerts

Shows real-time warning messages like "No Movement" or "High Heart Rate." Alerts help caregivers take immediate action to ensure patient safety.



9. Alerts Panel

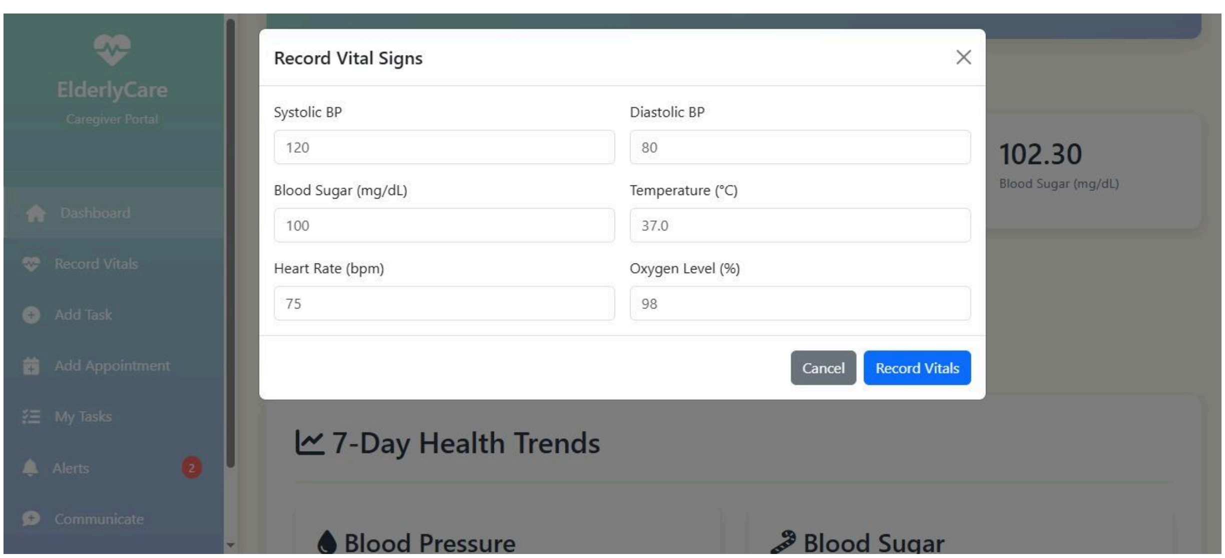
A dedicated place to review all past and recent alerts tied to the patient.



Data Entry & Communication

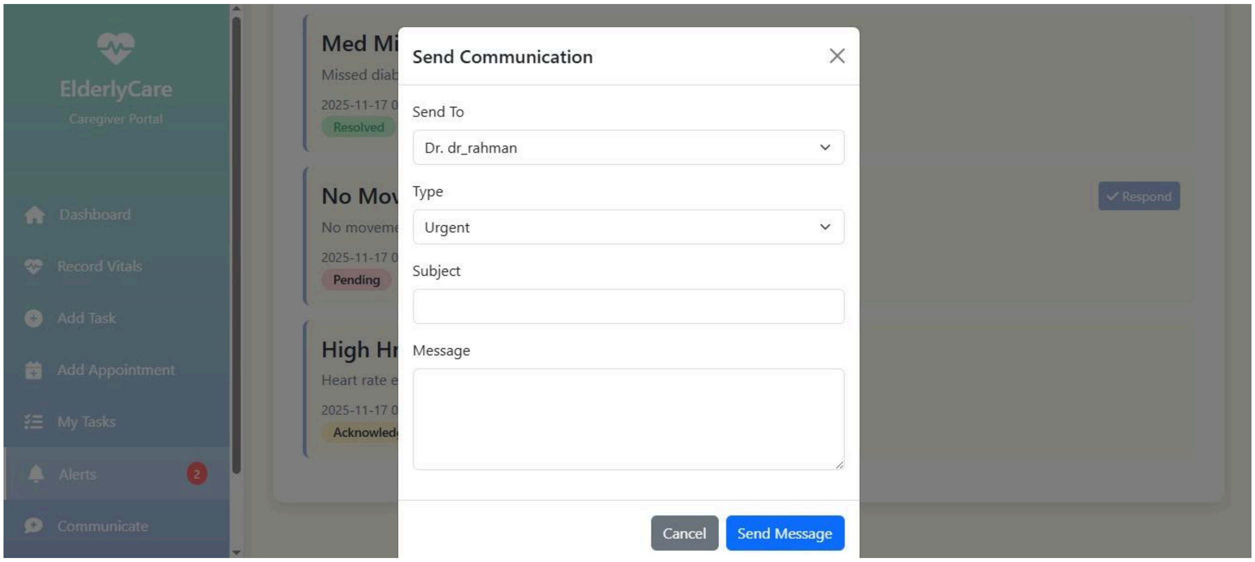
10. Record Vital Signs

A quick-entry form where caregivers manually record BP, sugar, temperature, heart rate, and symptoms—useful during home visits.



11. Send Communication

Allows sending short, direct messages (such as updates or warnings) to family members or doctors.



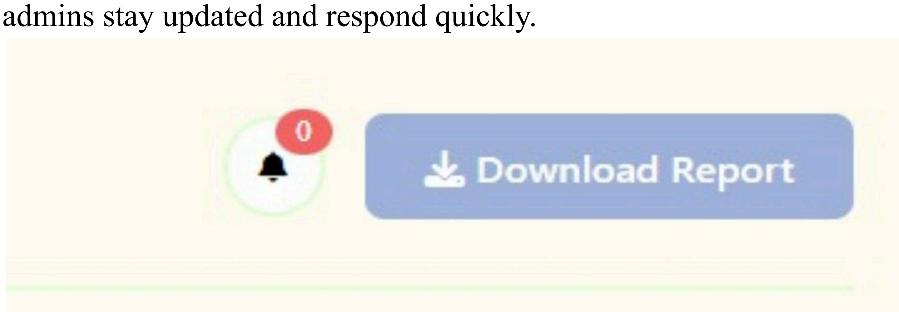
11.Family Dashboard

Dashboard Overview Page

A clear dashboard that shows the patient’s basic information along with real time health stats such as heart rate, blood pressure, blood sugar, and temperature. It also displays quick summaries for active alerts, upcoming appointments, medication adherence, and the last update time. A “Download Report” button is available for easy record keeping.

Recent Alerts Section

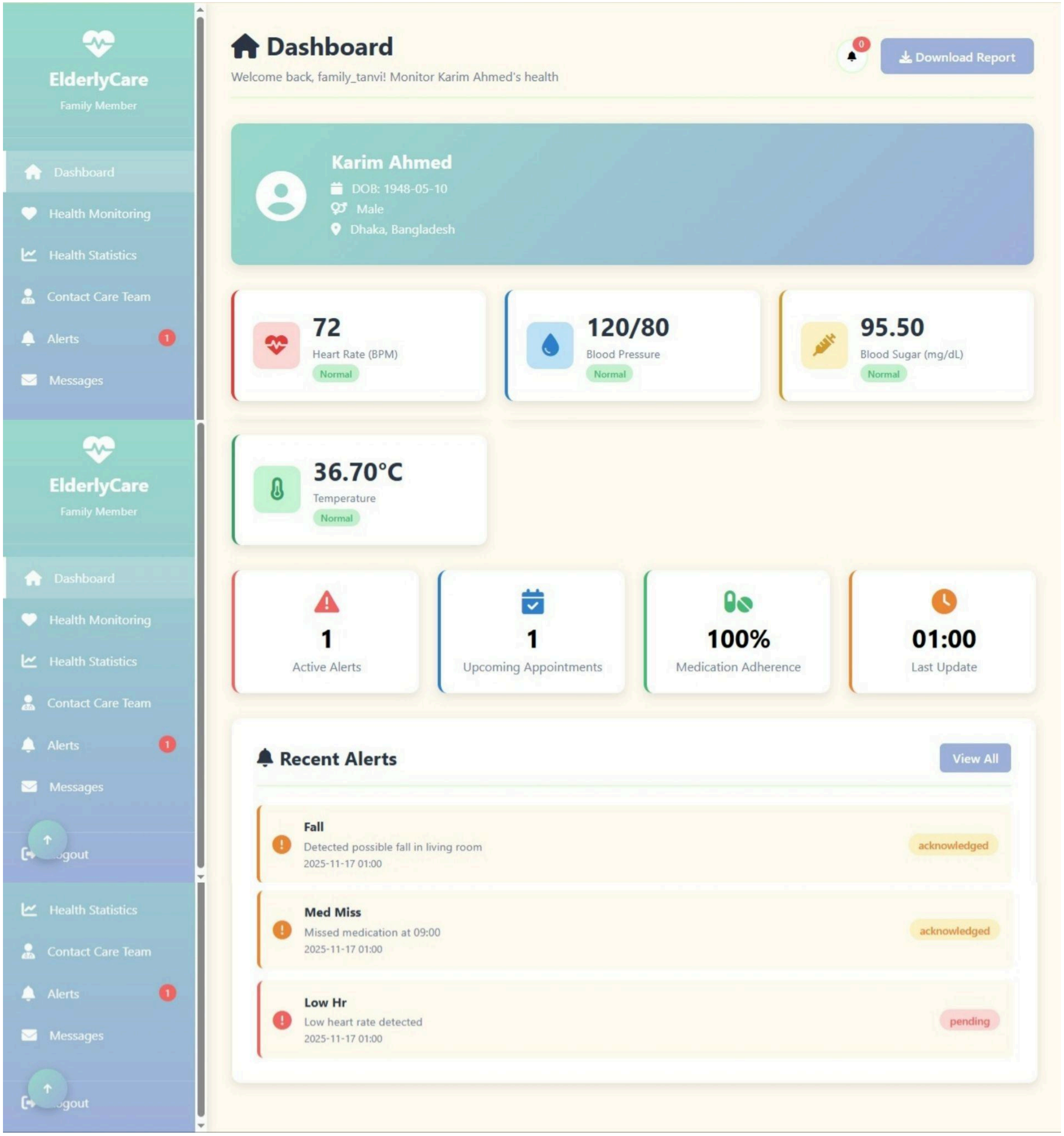
A dedicated section listing the latest health alerts, including issues like falls, missed medication, or abnormal readings. Each alert shows the time detected and its status, helping caregivers or admins stay updated and respond quickly.



Health Report - Karim Ahmed

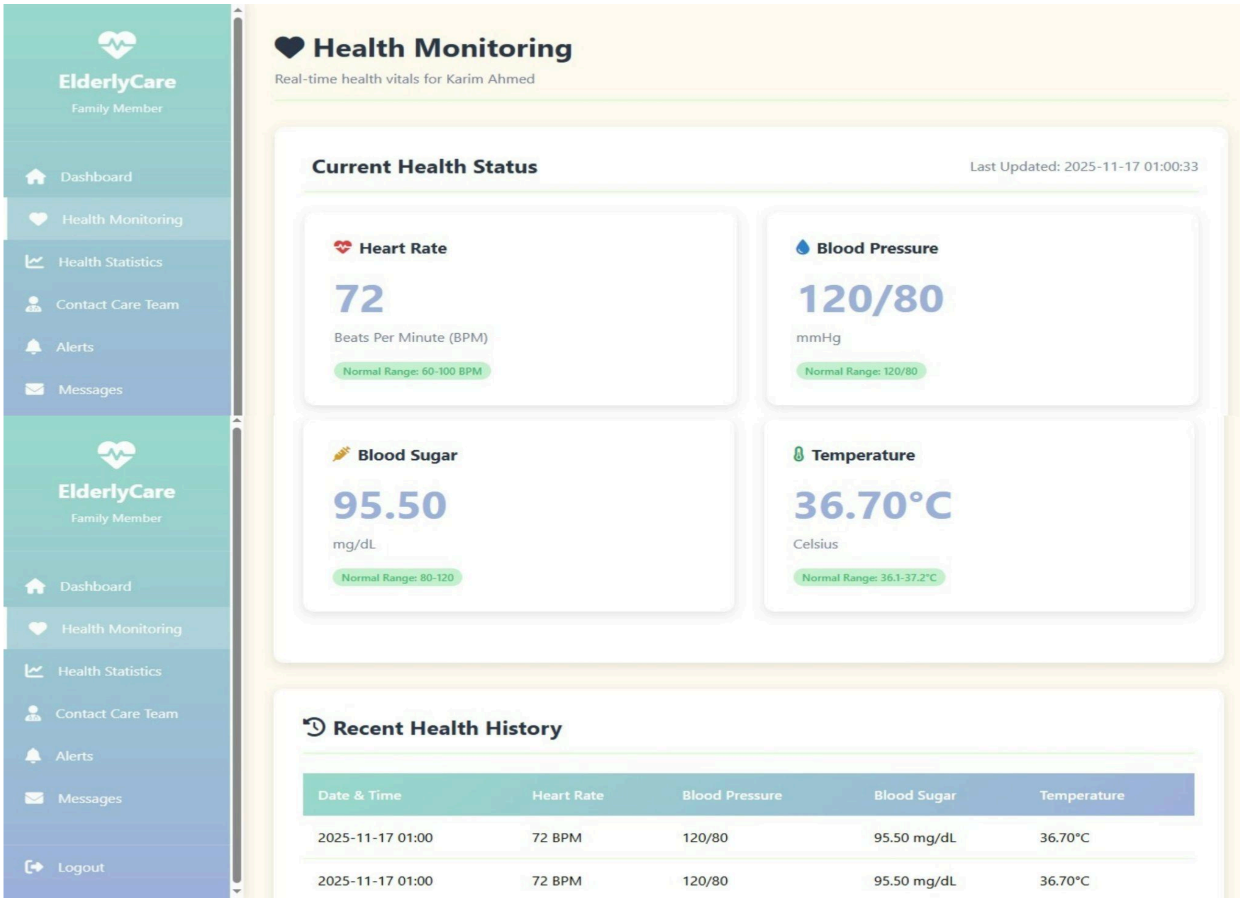
Generated on: 2025-11-21 13:26:51

Date	Heart Rate	Blood Pressure	Blood Sugar	Temperature
2025-11-17	72	120/80	95.50	36.70
2025-11-17	72	120/80	95.50	36.70



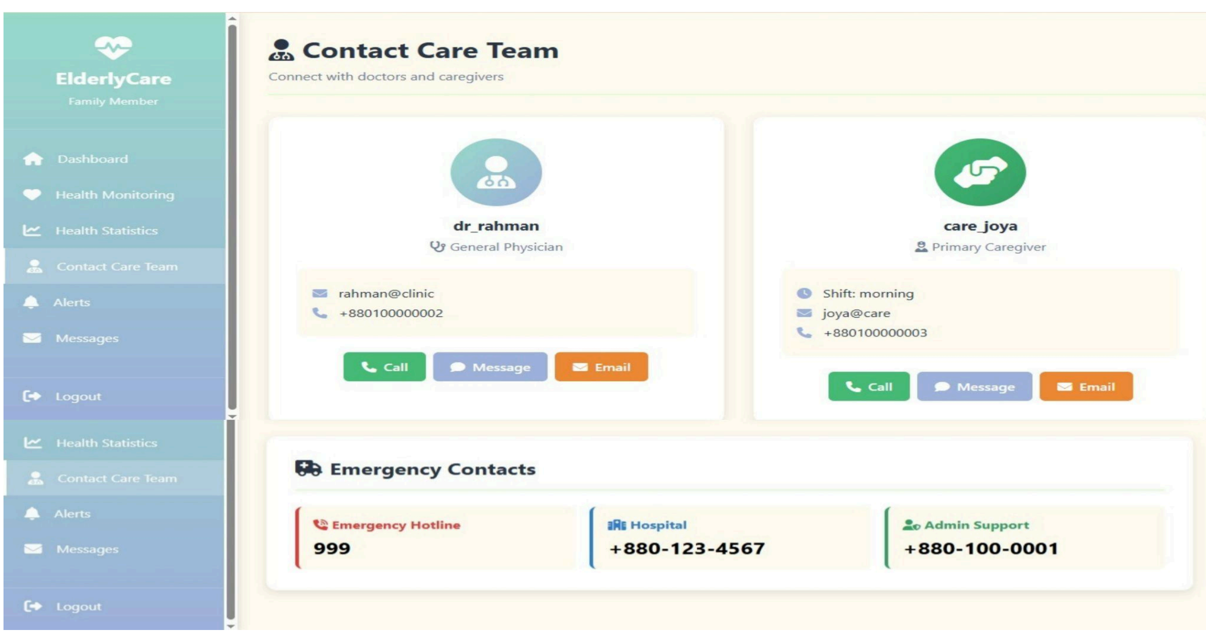
Health Monitoring Page

A simple overview of the patient's current vitals, including heart rate, blood pressure, blood sugar, and temperature. It also shows the latest update time and a table of recent health readings for quick review.



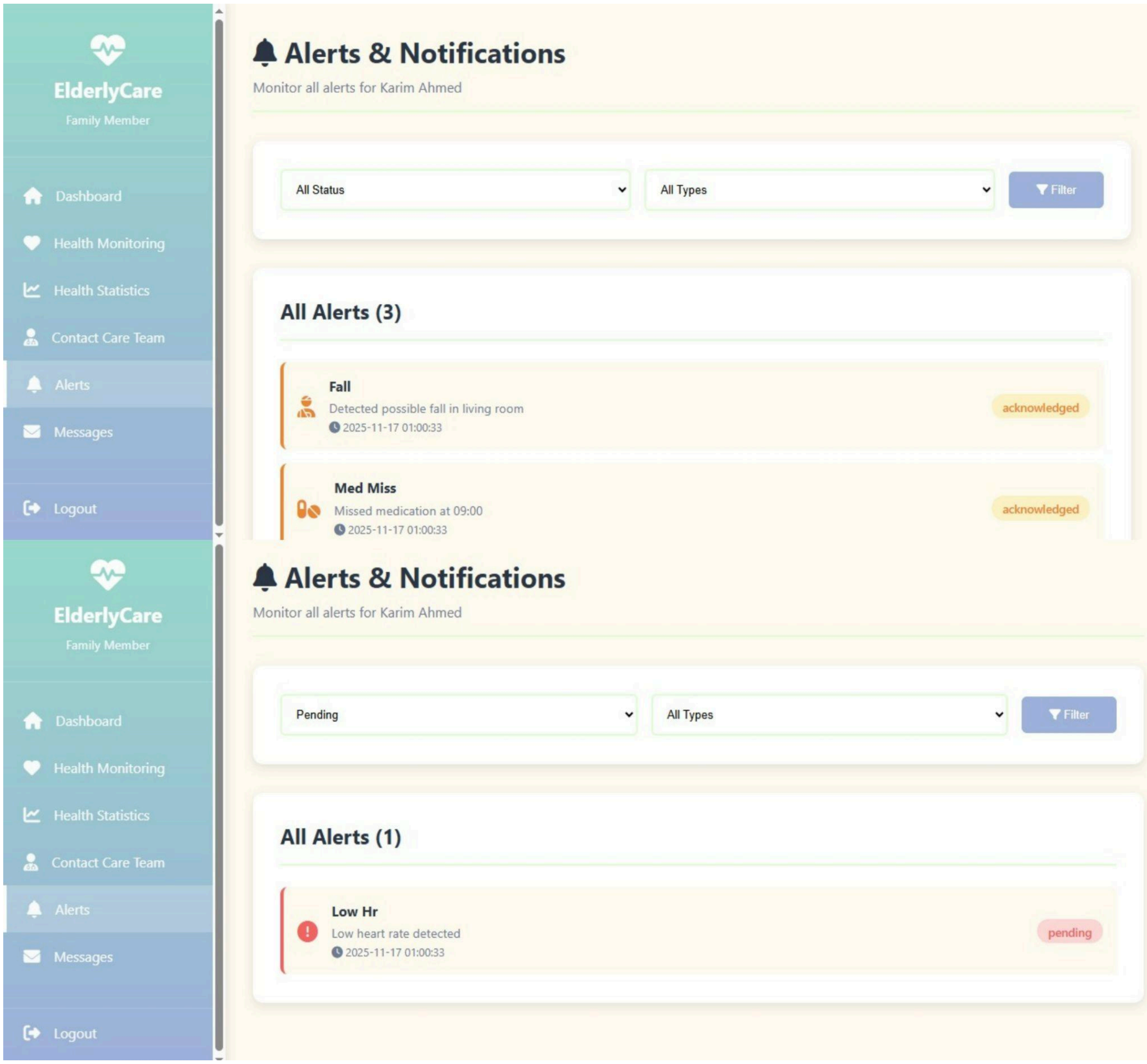
Health Statistics Page & Contact Care Team Page

Simple charts that help family members understand health trends over time, see what types of alerts happen most often, and identify when issues typically occur throughout the day.A convenient place to view and reach the assigned doctor and caregiver. Includes options to call, message, or email them, along with clearly listed emergency contact numbers.



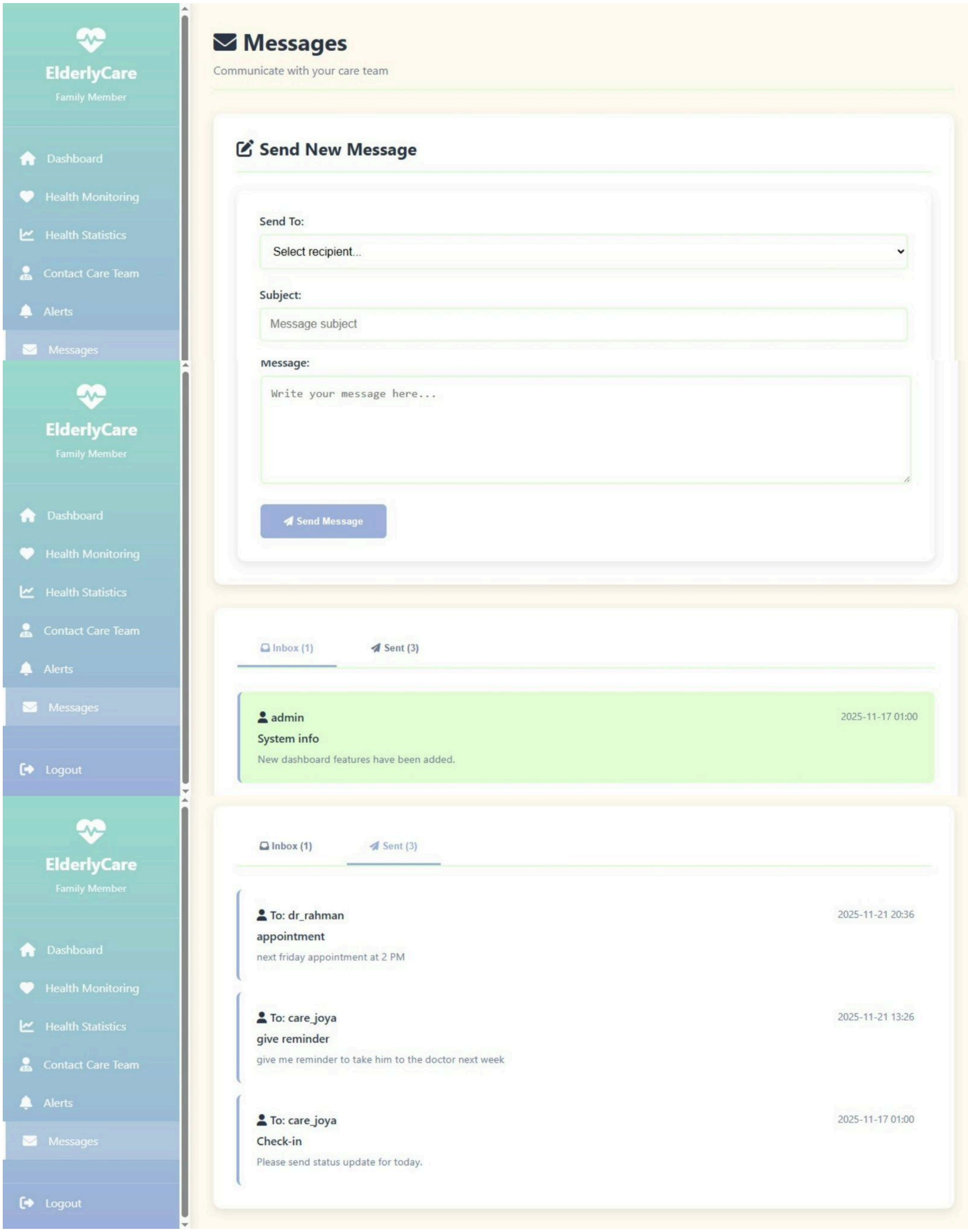
Alerts & Notifications Page

A clean list of recent alerts such as falls, missed medication, or unusual vitals. Each alert shows its time and status, helping families stay informed and respond quickly when needed.

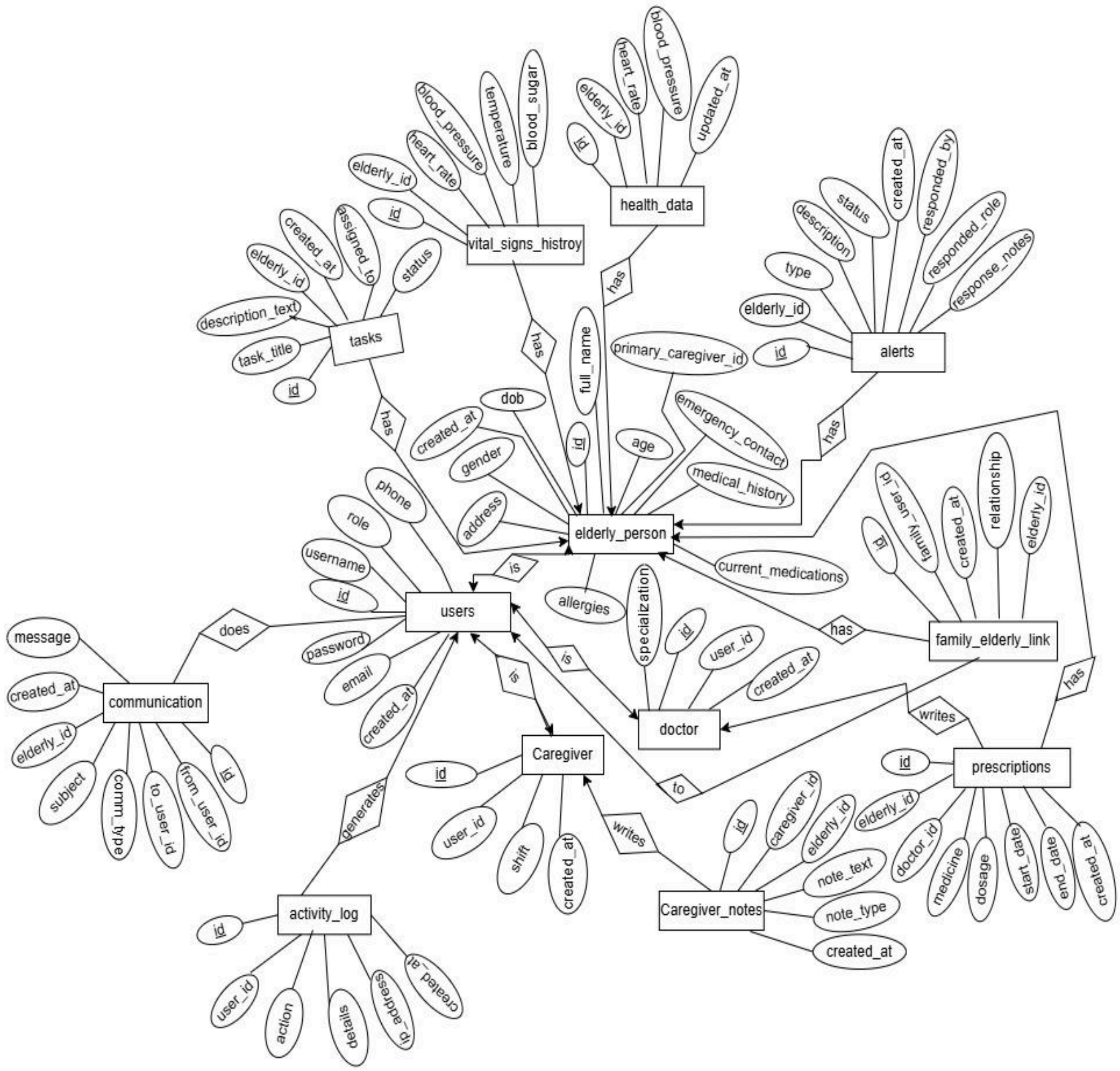


Messages Page

A simple communication page where family members can view and send messages directly to the care team, making it easy to stay connected and receive updates.


ElderlyCare
Family Member
Dashboard
Health Monitoring
Health Statistics
Contact Care Team
Alerts
Messages
Messages
Communicate with your care team
Send New Message
Send To:
Select recipient...
Subject:
Message subject
Message:
Write your message here...
Send Message
ElderlyCare
Family Member
Dashboard
Health Monitoring
Health Statistics
Contact Care Team
Alerts
Messages
Logout
Inbox (1)
Sent (3)
admin
2025-11-17 01:00
System info
New dashboard features have been added.
ElderlyCare
Family Member
Dashboard
Health Monitoring
Health Statistics
Contact Care Team
Alerts
Messages
Logout
Inbox (1)
Sent (3)
To: dr_rahman
2025-11-21 20:36
appointment
next friday appointment at 2 PM
To: care_joya
2025-11-21 13:26
give reminder
give me reminder to take him to the doctor next week
To: care_joya
2025-11-17 01:00
Check-in
Please send status update for today.

Entity Relationship Diagram



Narrative Description of ER Diagram

Entities:

1. Users

The **users** table is the central entity for all individuals who access the system, including administrators, doctors, caregivers, and family members. Each user is uniquely identified by *id* (primary key) and contains essential profile and login information such as username, email, password, phone number, role, and account creation date. This entity supports role-based access, where specialized entities like caregivers and doctors extend from it.

2. Elderly_Person

The **elderly_person** entity stores personal, medical, and contact information for each elderly individual receiving care. Each elderly person is uniquely identified by *id* and includes details like full name, DOB, age, gender, address, emergency contact, medical history, allergies, and current medications. A primary caregiver can also be assigned to each elderly person.

3. Caregiver

The **caregiver** entity represents users who provide daily assistance to elderly individuals. It includes a caregiver's unique *id*, linked *user_id*, shift information, and timestamps. Caregivers interact closely with elderly individuals through assigned tasks and note-taking.

4. Doctor

The **doctor** entity extends the users table and represents medical professionals who diagnose, prescribe medication, and review caregiving notes. Doctors have their own *id* linked to a user account.

5. Family_Elderly_Link

This table links family members (who are users) to elderly individuals. It represents their relationship type (e.g., daughter, son, spouse) and stores timestamps for when the connection was created.

6. Health_Data

The **health_data** entity stores the most recent vital readings of an elderly person, such as blood pressure, heart rate, blood sugar, and temperature. It captures real-time monitoring data with timestamps.

7. Vital_Signs_History

This entity keeps historical logs of vital signs for an elderly individual. Each record includes readings and a timestamp for when the vitals were captured.

8. Tasks

The **tasks** entity defines responsibilities assigned to caregivers. Each task includes a title, description, status, assigned caregiver, and the elderly person it belongs to.

9. Alerts

The **alerts** entity stores notifications regarding elderly individuals, such as health abnormalities or safety issues. Each alert includes a type, description, status, response notes, and who responded.

10. Prescriptions

The **prescriptions** table stores medication details written for an elderly person by a doctor, including dosage, type, start date, and end date.

11. Caregiver_Notes

The **caregiver_notes** entity contains text notes recorded by caregivers about observations, routines, or health changes of elderly persons.

12. Communication

Communication allows messaging between users regarding elderly persons. Each record includes sender, receiver, subject, message text, and timestamp.

13. Activity_Log

The activity_log records different actions performed in the system, such as updates, alerts acknowledged, task completion, and more. It stores user actions with descriptions and timestamps.

Relationships:

1. Users — Caregiver (is)

The *is* relationship connects the **users** table to the **caregiver** table. It indicates that every caregiver in the system must first be a registered user. This forms a one-to-one extension, where a user account provides authentication details while the caregiver table stores role-specific information such as shift and assigned elderly persons. Thus, a caregiver *is* always a user, and their identity is inherited from the users entity.

2. Users — Doctor (is)

The *is* relationship also links **users** to **doctor**, showing that each doctor is fundamentally a user of the system. Through this mapping, the doctor inherits their login credentials and profile information from the users table, while their professional role and responsibilities are handled separately in the doctor entity. This maintains centralized authentication while supporting specialized behavior.

3. Users — Communication (does)

The *does* relationship describes how **users** participate in communication activities. Each user may send or receive messages, and these interactions are recorded in the communication table. This relationship highlights that communication is a user-driven activity, allowing users to exchange information regarding elderly individuals' care, issues, and updates.

4. Users — Activity_Log (generates)

The *generates* relationship connects **users** to the **activity_log**, stating that all system actions originate from users. Whenever a user performs an operation—such as updating a profile, responding to alerts, or completing tasks—the action is captured in an activity log entry. This enables accountability, auditing, and tracking of user behaviors throughout the system.

5. Elderly_Person — Health_Data (has)

The *has* relationship between **elderly_person** and **health_data** signifies that each elderly individual has a corresponding set of real-time health measurements. This includes vitals such as blood pressure, sugar levels, temperature, and heart rate. The relationship ensures that every elderly person's current health status is continuously monitored and stored.

6. Elderly_Person — Vital_Signs_History (has)

The *has* relationship indicates that an **elderly_person** can have multiple historical vital readings stored in **vital_signs_history**. This enables long-term medical analysis by keeping records of changes in health metrics over time, allowing caregivers and doctors to monitor trends and identify abnormal patterns.

7. Elderly_Person — Tasks (has)

Through the *has* relationship, each **elderly_person** may have several caregiving tasks assigned to them. These tasks describe specific activities required for their care, such as routine checkups, medication reminders, or assistance activities. This ensures that all duties related to an elderly individual are clearly organized and assigned.

8. Caregiver — Tasks (has)

The *has* relationship also exists between **caregiver** and **tasks**, meaning a caregiver manages multiple tasks assigned to them. This relationship identifies the caregiver responsible for performing each task and ensures accountability by linking tasks to the correct caregiver.

9. Elderly_Person — Alerts (has)

The *has* relationship between **elderly_person** and **alerts** indicates that alerts are generated specifically for an elderly individual. These alerts may concern abnormal health readings, emergency situations, or safety warnings. The relationship ensures that each alert is associated with the correct elderly individual and can be managed accordingly.

10. Alerts — User (responded_by / responded_role)

Alerts include information about **which user responded** and in **what role**. This defines a relationship where users—acting as caregivers, doctors, or family—may respond to alerts. The response details are stored in the alert, ensuring transparency about how alerts were managed.

11. Doctor — Prescriptions (writes)

The *writes* relationship links **doctors** to **prescriptions**, indicating that only doctors are authorized to issue prescriptions for elderly individuals. Each prescription records the doctor responsible for prescribing medication, reinforcing medical accountability and ensuring proper authorization.

12. Elderly_Person — Prescriptions (has)

The *has* relationship shows that each **elderly_person** can have multiple prescriptions. These contain important medical instructions such as dosage, start dates, and end dates. This ensures that every medication provided is linked back to the correct elderly individual.

13. Caregiver — Caregiver_Notes (writes)

The *writes* relationship between **caregiver** and **caregiver_notes** indicates that caregivers are responsible for documenting observations, updates, and daily care notes. These notes help maintain continuity of care by storing written reports linked directly to the caregiver.

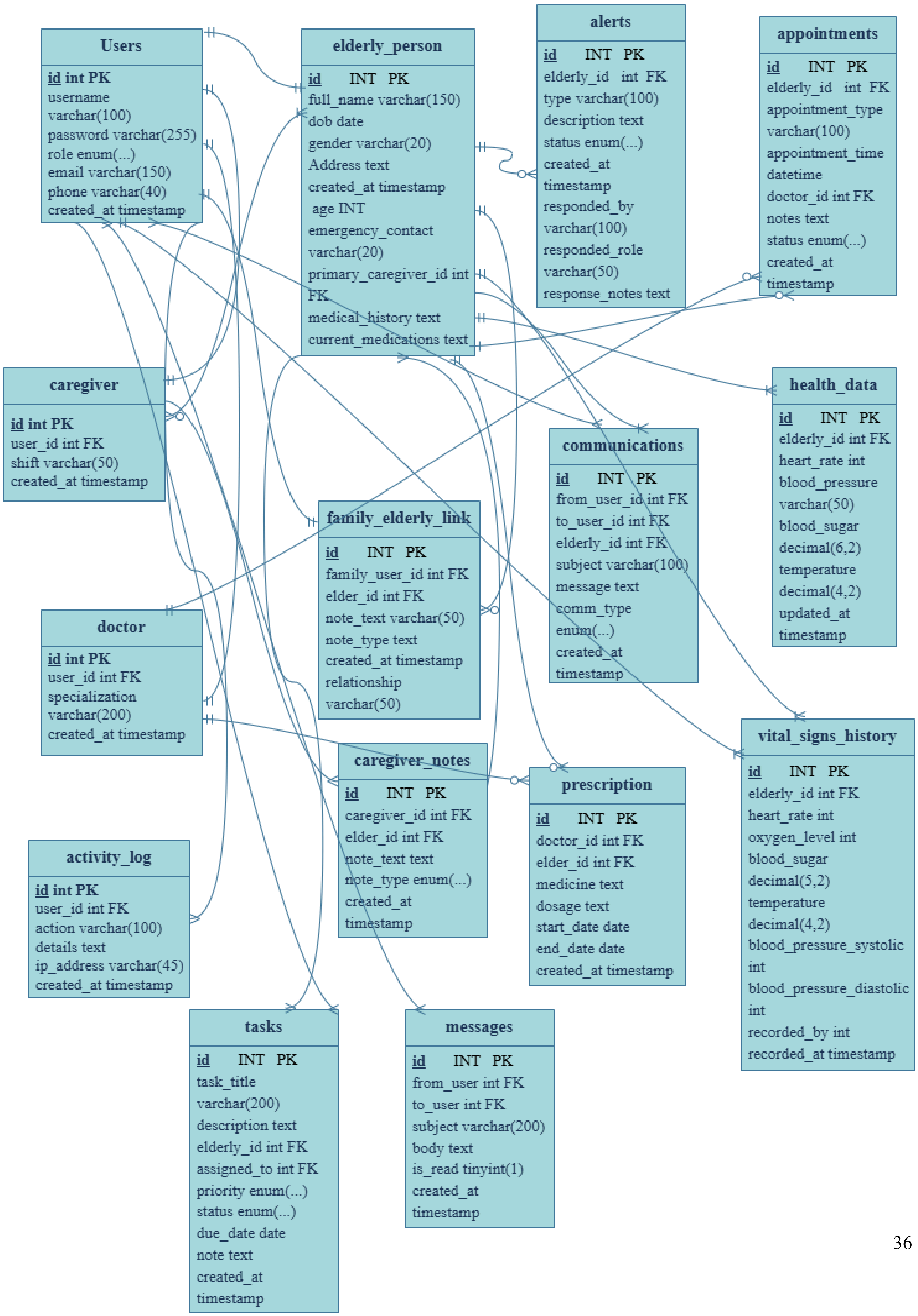
14. Elderly_Person — Caregiver_Notes (has)

Each elderly person *has* multiple notes written about them by caregivers. This relationship ensures that all daily observations, concerns, and updates are associated with the correct elderly individual, forming a reliable care history.

15. Family_Elderly_Link — Users & Elderly Person (has)

Through the *has* relationship, an elderly person can have multiple family members connected to them, and each family member can be associated with one or more elderly individuals. This linking table stores the relationship type and ensures that family members are correctly tied to the elderly for updates, permissions, and notifications.

Schema refinement



Integrity Constraints

Constraint Type	**Table**	**Column(s)**	**Description**
Primary Key	users	id	Unique identifier for each user
Unique Key	users	username	Ensures username is unique
Primary Key	elderly_person	id	Unique identifier for each elderly person
Foreign Key	elderly_person	primary_caregiver_id	References caregiver(id)
Primary Key	caregiver	id	Unique identifier for each caregiver
Foreign Key	caregiver	user_id	References users(id)
Primary Key	doctor	id	Unique identifier for each doctor
Foreign Key	doctor	user_id	References users(id)
Primary Key	family_elderly_link	id	Unique unique identifier for each link
Foreign Key	family_elderly_link	family_user_id	References users(id)
Foreign Key	family_elderly_link	elderly_id	References elderly_person(id)
Primary Key	alerts	id	Unique identifier for each alert
Foreign Key	alerts	elderly_id	References elderly_person(id)
Primary Key	appointments	id	Unique identifier for each appointment
Foreign Key	appointments	elderly_id	References elderly_person(id)

Foreign Key	appointments	doctor_id	References doctor(id)
Primary Key	prescriptions	id	Unique identifier for each prescription
Foreign Key	prescriptions	doctor_id	References doctor(id)
Foreign Key	prescriptions	elderly_id	References elderly_person(id)
Primary Key	communications	id	Unique identifier for each communication
Foreign Key	communications	from_user_id	References users(id)
Foreign Key	communications	to_user_id	References users(id)
Foreign Key	communications	elderly_id	References elderly_person(id)
Primary Key	caregiver_notes	id	Unique identifier for each caregiver note
Foreign Key	caregiver_notes	caregiver_id	References caregiver(id)
Foreign Key	caregiver_notes	elderly_id	References elderly_person(id)
Primary Key	health_data	id	Unique identifier for each health data entry
Foreign Key	health_data	elderly_id	References elderly_person(id)
Primary Key	messages	id	Unique identifier for each messages entry
Foreign Key	messages	from_user	References users(id)
Foreign Key	messages	to_user	References users(id)

Primary Key	vital_signs_history	id	Unique identifier for each vital entry
Foreign Key	vital_signs_history	elderly_id	References elderly_person(id)
Foreign Key	vital_signs_history	recorded_by	References users(id)
Primary Key	tasks	id	Unique identifier for each task
Foreign Key	tasks	elderly_id	References elderly_person(id)
Foreign Key	tasks	assigned_to	References users(id)
Primary Key	activity_log	id	Unique identifier for each log entry
Foreign Key	activity_log	user_id	References users(id)

Normalization

1st Normal Form (1NF)

The given relations are in 1st Normal Form because it meets the following criteria:

1. All cells contain atomic (indivisible) values.

2. Entries in each column are of the same type.

3. Each tuple is unique; there are no duplicate rows.

4. Column names are unique.

2nd Normal Form (2NF)

The given relations are in 2nd Normal Form because it satisfies these conditions:

1. The table is already in 1st Normal Form.

2. There is no partial dependency; no proper subset of a candidate key can uniquely identify a non-prime attribute.

3. All non-key attributes are fully dependent on the entire primary key.

3rd Normal Form (3NF)

The given relations are in 3rd Normal Form because it meets the following requirements:

1. The tables are in 2nd Normal Form.

2. There is no transitive dependency; no non-prime attribute can determine another non prime attribute.

3. For each functional dependency, either the left-hand side is a super key, or the right-hand side is a non-prime attribute.

4. All columns are determined only by the primary key and no other column.

Normalization Check

Table - Elderly_Person

```
mysql> select * from elderly_person;
+----+-------------+------------+--------+------------------------+---------------------+---------------------+------+-------------------+------------------+-----------+---------------------+
| id | full_name   | dob        | gender | address                | primary_caregiver_id | created_at          | age  | emergency_contact | medical_history  | allergies | current_medications |
+----+-------------+------------+--------+------------------------+---------------------+---------------------+------+-------------------+------------------+-----------+---------------------+
|  1 | Karim Ahmed | 1948-05-10 | Male   | Dhaka, Bangladesh      |                    3 | 2025-11-17 01:00:33 |   77 | 01712345678       | No major issues  | None      | Daily vitamins      |
|  2 | Jahan Begum | 1945-03-12 | Female | Chittagong, Bangladesh |                    8 | 2025-11-17 01:00:33 |   80 | 01712345678       | No major issues  | None      | Daily vitamins      |
|  3 | Mina Khatun | 1952-11-02 | Female | Khulna, Bangladesh     |                    9 | 2025-11-17 01:00:33 |   73 | 01712345678       | No major issues  | None      | Daily vitamins      |
|  4 | Hasan Ali   | 1940-01-22 | Male   | Rajshahi, Bangladesh   |                    9 | 2025-11-17 01:00:33 |   85 | 01712345678       | No major issues  | None      | Daily vitamins      |
+----+-------------+------------+--------+------------------------+---------------------+---------------------+------+-------------------+------------------+-----------+---------------------+
4 rows in set (0.00 sec)

mysql> describe elderly_person;
+----------------------+--------------+------+-----+-------------------+-------------------+
| Field                | Type         | Null | Key | Default           | Extra             |
+----------------------+--------------+------+-----+-------------------+-------------------+
| id                   | int          | NO   | PRI | NULL              | auto_increment    |
| full_name            | varchar(150) | NO   |     | NULL              |                   |
| dob                  | date         | YES  |     | NULL              |                   |
| gender               | varchar(20)  | YES  |     | NULL              |                   |
| address              | text         | YES  |     | NULL              |                   |
| primary_caregiver_id | int          | YES  | MUL | NULL              |                   |
| created_at           | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| age                  | int          | YES  |     | NULL              |                   |
| emergency_contact    | varchar(20)  | YES  |     | NULL              |                   |
| medical_history      | text         | YES  |     | NULL              |                   |
| allergies            | text         | YES  |     | NULL              |                   |
| current_medications  | text         | YES  |     | NULL              |                   |
+----------------------+--------------+------+-----+-------------------+-------------------+
12 rows in set (0.07 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(full_name,dob etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table - Activity_Log

```
mysql>  describe  activity_log;
+------------+--------------+------+-----+-------------------+-------------------+
| Field      | Type         | Null | Key | Default           | Extra             |
+------------+--------------+------+-----+-------------------+-------------------+
| id         | int          | NO   | PRI | NULL              | auto_increment    |
| user_id    | int          | NO   | MUL | NULL              |                   |
| action     | varchar(100) | NO   |     | NULL              |                   |
| details    | text         | YES  |     | NULL              |                   |
| ip_address | varchar(45)  | YES  |     | NULL              |                   |
| created_at | timestamp    | YES  | MUL | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+--------------+------+-----+-------------------+-------------------+
6 rows in set (0.00 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(action,details etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table - Alerts

```
mysql> describe  alerts;
+----------------+----------------------------------------------+------+-----+-------------------+-------------------+
| Field          | Type                                         | Null | Key | Default           | Extra             |
+----------------+----------------------------------------------+------+-----+-------------------+-------------------+
| id             | int                                          | NO   | PRI | NULL              | auto_increment    |
| elderly_id     | int                                          | YES  | MUL | NULL              |                   |
| type           | varchar(100)                                 | YES  |     | NULL              |                   |
| description    | text                                         | YES  |     | NULL              |                   |
| status         | enum('pending','acknowledged','resolved')    | YES  |     | pending           |                   |
| created_at     | timestamp                                    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| responded_by   | varchar(100)                                 | YES  |     | NULL              |                   |
| responded_role | varchar(50)                                  | YES  |     | NULL              |                   |
| response_notes | text                                         | YES  |     | NULL              |                   |
+----------------+----------------------------------------------+------+-----+-------------------+-------------------+
9 rows in set (0.07 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(type,description etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table - Appointments

```
mysql> describe appointments;
+------------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| Field            | Type                                          | Null | Key | Default           | Extra             |
+------------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| id               | int                                           | NO   | PRI | NULL              | auto_increment    |
| elderly_id       | int                                           | NO   | MUL | NULL              |                   |
| appointment_type | varchar(100)                                  | NO   |     | NULL              |                   |
| appointment_time | datetime                                      | NO   | MUL | NULL              |                   |
| doctor_id        | int                                           | YES  | MUL | NULL              |                   |
| status           | enum('scheduled','completed','cancelled')     | YES  | MUL | scheduled         |                   |
| notes            | text                                          | YES  |     | NULL              |                   |
| created_at       | timestamp                                     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+-----------------------------------------------+------+-----+-------------------+-------------------+
8 rows in set (0.01 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(status,notes etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table - Caregiver

```
mysql> describe caregiver;
+------------+-------------+------+-----+-------------------+-------------------+
| Field      | Type        | Null | Key | Default           | Extra             |
+------------+-------------+------+-----+-------------------+-------------------+
| id         | int         | NO   | PRI | NULL              | auto_increment    |
| user_id    | int         | NO   | MUL | NULL              |                   |
| shift      | varchar(50) | YES  |     | NULL              |                   |
| created_at | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+-------------+------+-----+-------------------+-------------------+
4 rows in set (0.01 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(shift) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table- Caregiver_Notes

```
mysql> describe caregiver_notes;
+--------------+-----------------------------------------------------+------+-----+-------------------+-------------------+
| Field        | Type                                                | Null | Key | Default           | Extra             |
+--------------+-----------------------------------------------------+------+-----+-------------------+-------------------+
| id           | int                                                 | NO   | PRI | NULL              | auto_increment    |
| caregiver_id | int                                                 | NO   | MUL | NULL              |                   |
| elderly_id   | int                                                 | NO   | MUL | NULL              |                   |
| note_text    | text                                                | NO   |     | NULL              |                   |
| note_type    | enum('general','medical','behavioral','dietary')    | YES  |     | general           |                   |
| created_at   | timestamp                                           | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------+-----------------------------------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.01 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(note_text,note_type etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table - Communications

```
mysql> describe communications;
+--------------+-------------------------------------------------------+------+-----+-------------------+-------------------+
| Field        | Type                                                  | Null | Key | Default           | Extra             |
+--------------+-------------------------------------------------------+------+-----+-------------------+-------------------+
| id           | int                                                   | NO   | PRI | NULL              | auto_increment    |
| from_user_id | int                                                   | NO   | MUL | NULL              |                   |
| to_user_id   | int                                                   | NO   | MUL | NULL              |                   |
| elderly_id   | int                                                   | YES  | MUL | NULL              |                   |
| subject      | varchar(200)                                          | YES  |     | NULL              |                   |
| message      | text                                                  | NO   |     | NULL              |                   |
| comm_type    | enum('family_notify','doctor_contact','urgent','routine') | YES  |     | routine           |                   |
| is_read      | tinyint(1)                                            | YES  |     | 0                 |                   |
| created_at   | timestamp                                             | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------+-------------------------------------------------------+------+-----+-------------------+-------------------+
9 rows in set (0.00 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(subject,is_read etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table- Doctor

```
mysql> describe  doctor;
+----------------+--------------+------+-----+-------------------+-------------------+
| Field          | Type         | Null | Key | Default           | Extra             |
+----------------+--------------+------+-----+-------------------+-------------------+
| id             | int          | NO   | PRI | NULL              | auto_increment    |
| user_id        | int          | NO   | MUL | NULL              |                   |
| specialization | varchar(100) | YES  |     | NULL              |                   |
| created_at     | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+--------------+------+-----+-------------------+-------------------+
4 rows in set (0.01 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields (specialization)depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table- Family_Elderly_Link

```
mysql>  describe  family_elderly_link;
+----------------+-------------+------+-----+-------------------+-------------------+
| Field          | Type        | Null | Key | Default           | Extra             |
+----------------+-------------+------+-----+-------------------+-------------------+
| id             | int         | NO   | PRI | NULL              | auto_increment    |
| family_user_id | int         | NO   | MUL | NULL              |                   |
| elderly_id     | int         | NO   | MUL | NULL              |                   |
| relationship   | varchar(50) | YES  |     | NULL              |                   |
| created_at     | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+-------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(relationship) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table- Health_Data

```
mysql> describe health_data;
+----------------+--------------+------+-----+-------------------+-------------------+
| Field          | Type         | Null | Key | Default           | Extra             |
+----------------+--------------+------+-----+-------------------+-------------------+
| id             | int          | NO   | PRI | NULL              | auto_increment    |
| elderly_id     | int          | NO   | MUL | NULL              |                   |
| heart_rate     | int          | YES  |     | NULL              |                   |
| blood_pressure | varchar(50)  | YES  |     | NULL              |                   |
| blood_sugar    | decimal(6,2) | YES  |     | NULL              |                   |
| temperature    | decimal(4,2) | YES  |     | NULL              |                   |
| updated_at     | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+--------------+------+-----+-------------------+-------------------+
7 rows in set (0.01 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(heart_rate,blood_sugar etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table- Messages

```
mysql>  describe messages;
+------------+--------------+------+-----+-------------------+-------------------+
| Field      | Type         | Null | Key | Default           | Extra             |
+------------+--------------+------+-----+-------------------+-------------------+
| id         | int          | NO   | PRI | NULL              | auto_increment    |
| from_user  | int          | YES  | MUL | NULL              |                   |
| to_user    | int          | YES  | MUL | NULL              |                   |
| subject    | varchar(200) | YES  |     | NULL              |                   |
| body       | text         | YES  |     | NULL              |                   |
| created_at | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| is_read    | tinyint(1)   | YES  |     | 0                 |                   |
+------------+--------------+------+-----+-------------------+-------------------+
7 rows in set (0.00 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(subject,body etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Tables -Prescriptions

```
mysql> describe prescriptions;
+------------+-----------+------+-----+-------------------+-------------------+
| Field      | Type      | Null | Key | Default           | Extra             |
+------------+-----------+------+-----+-------------------+-------------------+
| id         | int       | NO   | PRI | NULL              | auto_increment    |
| elderly_id | int       | YES  | MUL | NULL              |                   |
| doctor_id  | int       | YES  | MUL | NULL              |                   |
| medicine   | text      | YES  |     | NULL              |                   |
| dosage     | text      | YES  |     | NULL              |                   |
| start_date | date      | YES  |     | NULL              |                   |
| end_date   | date      | YES  |     | NULL              |                   |
| created_at | timestamp | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+-----------+------+-----+-------------------+-------------------+
8 rows in set (0.01 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(medicine,dosage etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table- Tasks

```
mysql> describe tasks;
+--------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| Field        | Type                                          | Null | Key | Default           | Extra             |
+--------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| id           | int                                           | NO   | PRI | NULL              | auto_increment    |
| task_title   | varchar(200)                                  | NO   |     | NULL              |                   |
| description  | text                                          | YES  |     | NULL              |                   |
| elderly_id   | int                                           | YES  | MUL | NULL              |                   |
| assigned_to  | int                                           | YES  | MUL | NULL              |                   |
| priority     | enum('low','medium','high')                   | YES  |     | medium            |                   |
| status       | enum('pending','in_progress','completed')     | YES  | MUL | pending           |                   |
| due_date     | date                                          | YES  | MUL | NULL              |                   |
| completed_at | timestamp                                     | YES  |     | NULL              |                   |
| notes        | text                                          | YES  |     | NULL              |                   |
| created_at   | timestamp                                     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------+-----------------------------------------------+------+-----+-------------------+-------------------+
11 rows in set (0.01 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(due_date,completed_at etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table- Users

```
mysql> describe  users;
+------------+-------------------------------------------------------+------+-----+-------------------+-------------------+
| Field      | Type                                                  | Null | Key | Default           | Extra             |
+------------+-------------------------------------------------------+------+-----+-------------------+-------------------+
| id         | int                                                   | NO   | PRI | NULL              | auto_increment    |
| username   | varchar(100)                                          | NO   | UNI | NULL              |                   |
| password   | varchar(255)                                          | NO   |     | NULL              |                   |
| role       | enum('admin','doctor','caregiver','family','elderly') | NO   |     | NULL              |                   |
| email      | varchar(150)                                          | YES  |     | NULL              |                   |
| phone      | varchar(40)                                           | YES  |     | NULL              |                   |
| created_at | timestamp                                             | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+-------------------------------------------------------+------+-----+-------------------+-------------------+
7 rows in set (0.00 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(role,email etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

Table- Vital_Signs_History

```
mysql> describe vital_signs_history;
+--------------------------+--------------+------+-----+-------------------+-------------------+
| Field                    | Type         | Null | Key | Default           | Extra             |
+--------------------------+--------------+------+-----+-------------------+-------------------+
| id                       | int          | NO   | PRI | NULL              | auto_increment    |
| elderly_id               | int          | NO   | MUL | NULL              |                   |
| blood_pressure_systolic  | int          | YES  |     | NULL              |                   |
| blood_pressure_diastolic | int          | YES  |     | NULL              |                   |
| blood_sugar              | decimal(5,2) | YES  |     | NULL              |                   |
| temperature              | decimal(4,2) | YES  |     | NULL              |                   |
| heart_rate               | int          | YES  |     | NULL              |                   |
| oxygen_level             | int          | YES  |     | NULL              |                   |
| recorded_by              | int          | YES  | MUL | NULL              |                   |
| recorded_at              | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------------------+--------------+------+-----+-------------------+-------------------+
10 rows in set (0.01 sec)
```

Normalization Stage	Explanation
First Normal Form	All fields contain atomic values. No repeating groups.
Second Normal Form	Here id is the PK. All other fields(oxygen_level,temperature etc) depend on it.
Third Normal Form	All attributes directly depend on id. No transitive dependencies.

The table is in 3rd Normal Form or 3NF.

SQL Statements

The necessary SQL commands required to create the database ‘tuition_finder_system with appropriate foreign keys are shown below:

Table- Users

```
CREATE TABLE users (

    id INT AUTO_INCREMENT PRIMARY KEY,

    username VARCHAR(100) NOT NULL UNIQUE,

    password VARCHAR(255) NOT NULL,

    role ENUM('admin','doctor','caregiver','family','elderly') NOT NULL,

    email VARCHAR(150),

    phone VARCHAR(40),

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

);
```

Table- elderly_person

```
CREATE TABLE elderly_person (
    id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(150) NOT NULL,
    dob DATE NOT NULL,
    gender VARCHAR(20) NOT NULL,
    address TEXT,
    primary_caregiver_id INT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    age INT,
    emergency_contact VARCHAR(20),
    medical_history TEXT,
    allergies TEXT,
    current_medications TEXT,

    FOREIGN KEY (primary_caregiver_id) REFERENCES caregiver(id)
);
```

Table- doctor

```
CREATE TABLE doctor (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    specialization VARCHAR(200),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
    FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Table- caregiver

```
CREATE TABLE caregiver (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    shift VARCHAR(50),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Table- family_elderly_link

```
CREATE TABLE family_elderly_link (
    id INT AUTO_INCREMENT PRIMARY KEY,
    family_user_id INT NOT NULL,
    elderly_id INT NOT NULL,
    relationship VARCHAR(50) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (family_user_id) REFERENCES users(id),
    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id)
);
```

Table- alerts

```
CREATE TABLE alerts (
    id INT AUTO_INCREMENT PRIMARY KEY,
    elderly_id INT NOT NULL,
    type VARCHAR(100),
    description TEXT,
    status ENUM('pending','acknowledged','resolved') DEFAULT 'pending',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    responded_by VARCHAR(100),
    responded_role VARCHAR(50),
    response_notes TEXT,

    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id)
);
```

Table- appointments

```
CREATE TABLE appointments (
    id INT AUTO_INCREMENT PRIMARY KEY,
    elderly_id INT NOT NULL,
    appointment_type VARCHAR(100) NOT NULL,
    appointment_time DATETIME NOT NULL,
    doctor_id INT,
    status ENUM('scheduled','completed','cancelled') DEFAULT 'scheduled',
    notes TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
    FOREIGN KEY (doctor_id) REFERENCES doctor(id),
    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id)
);
```

Table- caregiver_notes

```
CREATE TABLE caregiver_notes (
    id INT AUTO_INCREMENT PRIMARY KEY,
    caregiver_id INT NOT NULL,
    elderly_id INT NOT NULL,
    note_text TEXT NOT NULL,
    note_type ENUM('general','medical','behavioral','dietary') DEFAULT 'general',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (caregiver_id) REFERENCES caregiver(id),
    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id)
);
```

Table- communications

```
CREATE TABLE communications (
    id INT AUTO_INCREMENT PRIMARY KEY,
    from_user_id INT NOT NULL,
    to_user_id INT NOT NULL,
    elderly_id INT,
    subject VARCHAR(100),
```

```
    message TEXT,
    comm_type ENUM('family_notify','doctor_contact','urgent','routine') DEFAULT 'routine',
    is_read TINYINT(1) DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id),
    FOREIGN KEY (from_user_id) REFERENCES users(id),
    FOREIGN KEY (to_user_id) REFERENCES users(id)
);
```

Table- messages

```
CREATE TABLE messages (
    id INT AUTO_INCREMENT PRIMARY KEY,
    from_user INT,
    to_user INT,
    subject VARCHAR(200),
    body TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    is_read TINYINT(1),

    FOREIGN KEY (from_user) REFERENCES users(id),
    FOREIGN KEY (to_user) REFERENCES users(id)
);
```

Table- prescriptions

```
CREATE TABLE prescriptions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    elderly_id INT NOT NULL,
    doctor_id INT NOT NULL,
    medicine TEXT,
    dosage TEXT,
    start_date DATE,
    end_date DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (doctor_id) REFERENCES doctor(id),
    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id)
);
```

Table- tasks

```
CREATE TABLE tasks (
    id INT AUTO_INCREMENT PRIMARY KEY,
    task_title VARCHAR(200) NOT NULL,
    description TEXT,
    elderly_id INT,
    assigned_to INT,
    priority ENUM('low','medium','high') DEFAULT 'medium',
    status ENUM('pending','in_progress','completed') DEFAULT 'pending',
    due_date DATE,
```

```
    completed_at TIMESTAMP,
    notes TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (assigned_to) REFERENCES caregiver(id),
    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id)
);
```

Table- health_data

```
CREATE TABLE health_data (
    id INT AUTO_INCREMENT PRIMARY KEY,
    elderly_id INT NOT NULL,
    heart_rate INT,
    blood_pressure VARCHAR(50),
    blood_sugar DECIMAL(6,2),
    temperature DECIMAL(4,2),
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id)
);
```

Table- vital_signs_history

```
CREATE TABLE vital_signs_history (
    id INT AUTO_INCREMENT PRIMARY KEY,
    elderly_id INT NOT NULL,
```

```
    blood_pressure_systolic INT,
    blood_pressure_diastolic INT,
    blood_sugar DECIMAL(5,2),
    temperature DECIMAL(4,2),
    heart_rate INT,
    oxygen_level INT,
    recorded_by INT,
    recorded_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (elderly_id) REFERENCES elderly_person(id),
    FOREIGN KEY (recorded_by) REFERENCES users(id)
);
```

Table- activity_log

```
CREATE TABLE activity_log (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL,
    action VARCHAR(100) NOT NULL,
    details TEXT,
    ip_address VARCHAR(45),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Data Insertion in the Project Database

The elderly_care database was built step by step: first we created the database **"elderly_care"**. After that, we created the 15 tables in the correct order so foreign keys would work properly. We started by inserting the main people – users, doctors, caregivers and elderly persons as reference data. Once those were in place, we added appointments, prescriptions, daily vitals, health records, tasks, caregiver notes, messages, alerts and activity logs.

```
mysql> use elderly_care;
Database changed
mysql> show tables;
+------------------------+
| Tables_in_elderly_care |
+------------------------+
| activity_log           |
| alerts                 |
| appointments           |
| caregiver              |
| caregiver_notes        |
| communications         |
| doctor                 |
| elderly_person         |
| family_elderly_link    |
| health_data            |
| messages               |
| prescriptions          |
| tasks                  |
| users                  |
| vital_signs_history    |
+------------------------+
15 rows in set (0.01 sec)
```

Table - Elderly_Person

```
mysql> select * from elderly_person;
+----+-------------+------------+--------+-------------------------+----------------------+---------------------+------+-------------------+------------------+-----------+---------------------+
| id | full_name   | dob        | gender | address                 | primary_caregiver_id | created_at          | age  | emergency_contact | medical_history  | allergies | current_medications |
+----+-------------+------------+--------+-------------------------+----------------------+---------------------+------+-------------------+------------------+-----------+---------------------+
|  1 | Karim Ahmed | 1948-05-10 | Male   | Dhaka, Bangladesh       |                    3 | 2025-11-17 01:00:33 |   77 | 01712345678       | No major issues  | None      | Daily vitamins      |
|  2 | Jahan Begum | 1945-03-12 | Female | Chittagong, Bangladesh  |                    8 | 2025-11-17 01:00:33 |   80 | 01712345678       | No major issues  | None      | Daily vitamins      |
|  3 | Mina Khatun | 1952-11-02 | Female | Khulna, Bangladesh      |                    9 | 2025-11-17 01:00:33 |   73 | 01712345678       | No major issues  | None      | Daily vitamins      |
|  4 | Hasan Ali   | 1940-01-22 | Male   | Rajshahi, Bangladesh    |                    9 | 2025-11-17 01:00:33 |   85 | 01712345678       | No major issues  | None      | Daily vitamins      |
+----+-------------+------------+--------+-------------------------+----------------------+---------------------+------+-------------------+------------------+-----------+---------------------+
4 rows in set (0.00 sec)

mysql> describe elderly_person;
+----------------------+--------------+------+-----+-------------------+-------------------+
| Field                | Type         | Null | Key | Default           | Extra             |
+----------------------+--------------+------+-----+-------------------+-------------------+
| id                   | int          | NO   | PRI | NULL              | auto_increment    |
| full_name            | varchar(150) | NO   |     | NULL              |                   |
| dob                  | date         | YES  |     | NULL              |                   |
| gender               | varchar(20)  | YES  |     | NULL              |                   |
| address              | text         | YES  |     | NULL              |                   |
| primary_caregiver_id | int          | YES  | MUL | NULL              |                   |
| created_at           | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| age                  | int          | YES  |     | NULL              |                   |
| emergency_contact    | varchar(20)  | YES  |     | NULL              |                   |
| medical_history      | text         | YES  |     | NULL              |                   |
| allergies            | text         | YES  |     | NULL              |                   |
| current_medications  | text         | YES  |     | NULL              |                   |
+----------------------+--------------+------+-----+-------------------+-------------------+
12 rows in set (0.07 sec)
```

Table - Activity_Log

mysql> SELECT * FROM activity_log;

id	user_id	action	details	ip_address	created_at
1	1	Login	Successful login	192.168.1.100	2025-11-19 00:37:08
2	1	User Created	Created new doctor account	192.168.1.100	2025-11-19 00:37:08
3	2	Login	Successful login	192.168.1.101	2025-11-19 00:37:08
4	2	Updated Profile	Changed email address	192.168.1.101	2025-11-19 00:37:08
5	3	Login	Successful login	192.168.1.102	2025-11-19 00:37:08
6	3	Task Completed	Marked morning medication task as complete	192.168.1.102	2025-11-19 00:37:08
7	1	Alert Acknowledged	Acknowledged fall alert for Karim Ahmed	192.168.1.100	2025-11-18 23:37:11
8	6	Login	Successful login	192.168.1.103	2025-11-18 22:37:11
9	6	Prescription Added	Added new prescription for Jahan Begum	192.168.1.103	2025-11-18 22:37:11
10	8	Login	Successful login	192.168.1.104	2025-11-18 21:37:11
11	4	Login	Successful login	192.168.1.105	2025-11-18 19:37:11
12	4	Message Sent	Sent message to caregiver	192.168.1.105	2025-11-18 19:37:11
13	1	Login	Successful login	127.0.0.1	2025-11-19 10:44:28

Table - Alerts

mysql> SELECT * FROM alerts;

id	elderly_id	type	description	status	created_at	responded_by	responded_role	response_notes
1	1	fall	Detected possible fall in living room	acknowledged	2025-11-17 01:00:33	NULL	NULL	NULL
2	1	med_miss	Missed medication at 09:00	acknowledged	2025-11-17 01:00:33	NULL	NULL	NULL
3	1	low_hr	Low heart rate detected	pending	2025-11-17 01:00:33	NULL	NULL	NULL
4	2	high_bp	Blood pressure reached 180/110	pending	2025-11-17 01:00:33	NULL	NULL	NULL
5	2	fall	Fall detected in bathroom	acknowledged	2025-11-17 01:00:33	care_rahim	caregiver	NULL
6	3	low_oxygen	Oxygen level dropped to 88%	pending	2025-11-17 01:00:33	NULL	NULL	NULL
7	3	med_miss	Missed diabetes medication	resolved	2025-11-17 01:00:33	care_mitu	caregiver	NULL
8	4	no_movement	No movement detected for 45 mins	pending	2025-11-17 01:00:33	NULL	NULL	NULL
9	4	high_hr	Heart rate exceeded 130 BPM	acknowledged	2025-11-17 01:00:33	dr_kamal	doctor	NULL

9 rows in set (0.01 sec)

Table - Appointments

mysql> select * from appointments;

id	elderly_id	appointment_type	appointment_time	doctor_id	status	notes	created_at
1	1	Regular Checkup	2025-11-22 01:45:03	NULL	scheduled	Routine health check	2025-11-21 23:45:03
2	1	Physiotherapy	2025-11-22 04:45:03	NULL	scheduled	Weekly therapy	2025-11-21 23:45:03
3	1	Blood Test	2025-11-22 23:45:03	NULL	scheduled	Monthly blood work	2025-11-21 23:45:03
4	1	Dental Checkup	2025-11-24 23:45:03	NULL	scheduled	Dental check	2025-11-21 23:45:03
5	3	eye checkup	2025-11-22 18:10:00	NULL	scheduled	take previous record of prescription	2025-11-22 16:05:17

5 rows in set (0.01 sec)

Table - Caregiver

mysql> SELECT * FROM caregiver;

id	user_id	shift	created_at
1	3	morning	2025-11-17 01:00:33
2	8	evening	2025-11-17 01:00:33
3	9	night	2025-11-17 01:00:33

3 rows in set (0.01 sec)

Table - Caregiver_Notes

mysql> SELECT * FROM caregiver_notes;

id	caregiver_id	elderly_id	note_text	note_type	created_at
1	3	1	Patient doing well today. Ate full breakfast.	general	2025-11-19 15:35:25
2	3	2	Patient doing well today. Ate full breakfast.	general	2025-11-19 15:35:25
3	3	3	Patient doing well today. Ate full breakfast.	general	2025-11-19 15:35:25
4	9	3	missed last sunday's appointment	medical	2025-11-22 08:30:53

4 rows in set (0.04 sec)

Table - Communications

mysql> SELECT * FROM communications;

id	from_user_id	to_user_id	elderly_id	subject	message	comm_type	is_read	created_at
1	3	4	1	Daily Update	Karim had a good day today. All vital signs are normal.	family_notify	0	2025-11-19 15:35:28
2	9	2	3	appointment	miss her last sunday's appointment	doctor_contact	0	2025-11-22 08:32:45

2 rows in set (0.01 sec)

Table- Doctor

mysql> SELECT * FROM doctor;

id	user_id	specialization	created_at
1	2	General Physician	2025-11-17 01:00:33
2	6	Cardiologist	2025-11-17 01:00:33
3	7	Neurologist	2025-11-17 01:00:33

3 rows in set (0.01 sec)

Table- Family_Elderly_Link

mysql> SELECT * FROM family_elderly_link;

id	family_user_id	elderly_id	relationship	created_at
1	4	1	Son	2025-11-19 16:36:26
2	4	1	Son	2025-11-21 18:20:52

2 rows in set (0.01 sec)

Table- Health_Data

mysql> SELECT * FROM health_data;

id	elderly_id	heart_rate	blood_pressure	blood_sugar	temperature	updated_at
1	1	72	120/80	95.50	36.70	2025-11-17 01:00:33
2	2	88	140/90	120.20	37.10	2025-11-17 01:00:33
3	1	72	120/80	95.50	36.70	2025-11-17 01:00:33
4	2	88	140/90	120.20	37.10	2025-11-17 01:00:33
5	3	76	130/85	110.00	36.80	2025-11-17 01:00:33
6	4	95	150/95	105.70	37.30	2025-11-17 01:00:33
7	2	70	118/78	98.40	36.60	2025-11-17 01:00:33
8	3	82	135/88	115.00	36.90	2025-11-17 01:00:33
9	4	90	145/92	102.30	37.20	2025-11-17 01:00:39

9 rows in set (0.01 sec)

Table- Messages

mysql> SELECT * FROM messages;

id	from_user	to_user	subject	body	created_at	is_read
1	4	3	Check-in	Please send status update for today.	2025-11-17 01:00:33	0
2	1	2	System note	Monthly reports available.	2025-11-17 01:00:33	0
3	10	8	Request update	How is Jahan doing today?	2025-11-17 01:00:33	0
4	11	9	Medicine check	Please confirm if Mina took her medicine.	2025-11-17 01:00:33	0
5	6	9	Prescription update	Increase daily monitoring for Hasan	2025-11-17 01:00:33	0
6	12	8	Assistance needed	Please come to my room.	2025-11-17 01:00:33	0
7	1	4	System info	New dashboard features have been added.	2025-11-17 01:00:33	0
8	4	3	give reminder	give me reminder to take him to the doctor next week	2025-11-21 13:26:22	0
9	4	2	appointment	next friday appointment at 2 PM	2025-11-21 20:36:44	0

9 rows in set (0.02 sec)

Tables -Prescriptions

mysql> SELECT * FROM prescriptions;

id	elderly_id	doctor_id	medicine	dosage	start_date	end_date	created_at
1	1	1	Paracetamol 500mg	1 tab twice daily	2025-11-17	2025-11-24	2025-11-17 01:00:33
2	2	2	Amlodipine 5mg	1 tablet daily	2025-11-17	2025-12-17	2025-11-17 01:00:33
3	2	3	Vitamin B12	One capsule daily	2025-11-17	2026-02-15	2025-11-17 01:00:33
4	3	1	Metformin 500mg	Twice daily	2025-11-17	2026-01-16	2025-11-17 01:00:33
5	4	2	Aspirin 81mg	One daily	2025-11-17	2026-01-01	2025-11-17 01:00:33

5 rows in set (0.09 sec)

Table- Tasks

mysql> SELECT * FROM tasks;

id	task_title	description	elderly_id	assigned_to	priority	status	due_date	completed_at	notes	created_at
1	Morning Medication	Give morning medicines to patient	1	3	high	pending	2025-11-21	NULL	NULL	2025-11-21 23:45:03
2	Physical Therapy	Assist with daily physical therapy exercises	1	3	medium	pending	2025-11-21	NULL	NULL	2025-11-21 23:45:03
3	Check Vital Signs	Record blood pressure, temperature, heart rate	1	3	high	pending	2025-11-21	NULL	NULL	2025-11-21 23:45:03
4	Lunch Supervision	Ensure patient eats proper lunch	1	3	medium	pending	2025-11-21	NULL	NULL	2025-11-21 23:45:03
5	Evening Walk	Assist with evening walk	1	3	low	pending	2025-11-21	NULL	NULL	2025-11-21 23:45:03
6	check	evening meditation	3	9	medium	pending	2025-11-23	NULL	NULL	2025-11-22 14:04:05

6 rows in set (0.01 sec)

Table- Users

```
mysql> select * from users;
```

id	username	password	role	email	phone	created_at
1	admin	adminpass	admin	admin@local	+880100000001	2025-11-17 01:00:33
2	dr_rahman	drpass	doctor	rahman@clinic	+880100000002	2025-11-17 01:00:33
3	care_joya	carepass	caregiver	joya@care	+880100000003	2025-11-17 01:00:33
4	family_tanvi	familypass	family	tanvi@fam	+880100000004	2025-11-17 01:00:33
5	elder_karim	elderpass	elderly	karim@home	+880100000005	2025-11-17 01:00:33
6	dr_samina	pass123	doctor	samina@clinic	+880190000006	2025-11-17 01:00:33
7	dr_kamal	pass123	doctor	kamal@clinic	+880190000007	2025-11-17 01:00:33
8	care_rahim	carepass2	caregiver	rahim@care	+880190000008	2025-11-17 01:00:33
9	care_mitu	mitupass	caregiver	mitu@care	+880190000009	2025-11-17 01:00:33
10	family_ahmed	ahmedfam	family	ahmed@fam	+880190000010	2025-11-17 01:00:33
11	family_sara	sarapass	family	sara@fam	+880190000011	2025-11-17 01:00:33
12	elder_jahan	jahanpass	elderly	jahan@home	+880190000012	2025-11-17 01:00:33
13	elder_mina	minapass	elderly	mina@home	+880190000013	2025-11-17 01:00:33
14	elder_hasan	hasanpass	elderly	hasan@home	+880190000014	2025-11-17 01:00:33
16	karin_ahmed	password123	elderly	karin@example.com	01712345678	2025-11-28 01:27:05
17	mina_khatun	password123	elderly	mina@example.com	01712345679	2025-11-28 01:31:01

Table- Vital_Signs_History

```
mysql> SELECT * FROM vital_signs_history;
```

id	elderly_id	blood_pressure_systolic	blood_pressure_diastolic	blood_sugar	temperature	heart_rate	oxygen_level	recorded_by	recorded_at
1	4	139	83	94.20	37.30	77	96	3	2025-11-19 15:35:25
2	3	111	87	101.80	37.80	75	95	3	2025-11-13 15:35:25
3	2	110	77	119.10	37.30	83	96	3	2025-11-13 15:35:25
4	1	136	80	113.70	37.10	71	98	3	2025-11-16 15:35:25
5	4	115	73	103.40	36.80	66	98	3	2025-11-18 15:35:25
6	3	146	70	106.70	38.00	84	95	3	2025-11-15 15:35:25
7	2	115	87	125.70	37.80	85	98	3	2025-11-14 15:35:25
8	1	132	80	123.40	37.50	88	94	3	2025-11-15 15:35:25
9	4	120	75	116.40	37.20	74	94	3	2025-11-14 15:35:25
10	3	141	77	108.30	36.80	85	98	3	2025-11-15 15:35:25
11	2	149	70	100.00	36.70	90	99	3	2025-11-14 15:35:25
12	1	118	86	105.40	37.30	75	95	3	2025-11-18 15:35:25
13	4	121	86	101.50	37.90	90	96	3	2025-11-16 15:35:25
14	3	127	81	112.40	36.60	85	95	3	2025-11-13 15:35:25
15	2	119	74	110.00	37.70	78	99	3	2025-11-17 15:35:25
16	1	139	86	120.20	37.10	83	99	3	2025-11-13 15:35:25
17	4	145	80	92.10	37.50	67	96	3	2025-11-13 15:35:25
18	3	126	75	93.70	37.50	94	99	3	2025-11-14 15:35:25
19	2	146	76	125.60	37.20	86	95	3	2025-11-13 15:35:25
20	1	144	81	106.10	36.80	91	98	3	2025-11-18 15:35:25

```
20 rows in set (0.07 sec)
```

Complex Queries

1.Find doctors with appointments scheduled within 24 hours after an emergency (count per doctor)

Query:

SELECT d.id AS doctor_id, u.username AS doctor_user, COUNT(DISTINCT a.id) AS appts_within_24h

FROM alerts al

JOIN appointments a ON a.elderly_id = al.elderly_id

AND a.appointment_time BETWEEN al.created_at AND DATE_ADD(al.created_at, INTERVAL 24 HOUR)

JOIN doctor d ON d.id = a.doctor_id

JOIN users u ON u.id = d.user_id

WHERE al.type = 'emergency' AND a.doctor_id IS NOT NULL

GROUP BY d.id, u.username

ORDER BY appts_within_24h DESC;

2. Find family member engagement score: (family_notify read% over last 30 days) per family_user

Query:

```
WITH fam_comms AS (
  SELECT f.family_user_id, COUNT(*) total, SUM(c.is_read) read_count
  FROM family_elderly_link f
  LEFT JOIN communications c ON c.elderly_id = f.elderly_id AND c.comm_type='family_notify' AND c.created_at >= DATE_SUB(NOW(), INTERVAL 30 DAY)
  GROUP BY f.family_user_id
)
SELECT u.id, u.username,
  ROUND(100 * COALESCE(fc.read_count,0) / NULLIF(COALESCE(fc.total,0),0),2) AS pct_read
FROM users u
LEFT JOIN fam_comms fc ON fc.family_user_id = u.id
WHERE u.role = 'family'
ORDER BY pct_read DESC;
```

3. Find elderly who had an emergency alert but no urgent communication within 30 minutes

Query:

```
WITH comm_after AS (

  SELECT al.id AS alert_id, al.elderly_id, al.created_at AS alert_at,

        MIN(c.created_at) AS first_comm_at

  FROM alerts al

  LEFT JOIN communications c ON c.elderly_id = al.elderly_id AND c.created_at >=
al.created_at

  WHERE al.type = 'emergency'

  GROUP BY al.id, al.elderly_id, al.created_at

)

SELECT ca.alert_id, ca.elderly_id, e.full_name, ca.alert_at, ca.first_comm_at,

      CASE

        WHEN ca.first_comm_at IS NULL THEN 'no_comm'

        WHEN TIMESTAMPDIFF(MINUTE, ca.alert_at, ca.first_comm_at) > 30 THEN 'late'

        ELSE 'ok'

      END AS status

FROM comm_after ca

JOIN elderly_person e ON e.id = ca.elderly_id

WHERE CASE

       WHEN ca.first_comm_at IS NULL THEN 'no_comm'

       WHEN TIMESTAMPDIFF(MINUTE, ca.alert_at, ca.first_comm_at) > 30 THEN 'late'

       ELSE 'ok'

     END <> 'ok';
```

4.Find alerts that repeated (same type) for same elderly within 48 hours

Query:

```
SELECT a1.elderly_id, e.full_name, a1.type,
       a1.created_at AS first_time, a2.created_at AS repeat_time,
       TIMESTAMPDIFF(HOUR, a1.created_at, a2.created_at) AS hours_between
FROM alerts a1
JOIN alerts a2 ON a1.elderly_id = a2.elderly_id
  AND a1.type = a2.type
  AND a1.id < a2.id
  AND a2.created_at <= DATE_ADD(a1.created_at, INTERVAL 48 HOUR)
JOIN elderly_person e ON e.id = a1.elderly_id
ORDER BY a1.elderly_id, a1.created_at;
```

5. What is the average number of minutes between the creation of an alert and the caregiver's first communication regarding the same elderly individual?

Query:

```
WITH first_comm AS (
  SELECT al.id AS alert_id, al.elderly_id, al.created_at AS alert_at,
         MIN(c.created_at) AS first_comm_at
  FROM alerts al
  LEFT JOIN communications c
    ON c.elderly_id = al.elderly_id AND c.created_at >= al.created_at
  GROUP BY al.id, al.elderly_id, al.created_at
)
SELECT
```

CASE WHEN first_comm_at IS NULL THEN 'no_comm' ELSE 'commed' END AS comm_status,

COUNT(*) AS count_alerts,

ROUND(AVG(TIMESTAMPDIFF(MINUTE, alert_at, first_comm_at)),2) AS avg_minutes_to_first_comm

FROM first_comm

GROUP BY comm_status;

6.Which user recorded the most vitals in last 30 days (top 1)

Query:

SELECT u.id, u.username, COUNT(*) total_records

FROM vital_signs_history v

JOIN users u ON u.id = v.recorded_by

WHERE v.recorded_at >= DATE_SUB(NOW(), INTERVAL 30 DAY)

GROUP BY u.id

ORDER BY total_records DESC

LIMIT 1;

7.For each elderly find most recent appointment status and next scheduled appointment time.

Query:

WITH appt_rank AS (

SELECT a.*, ROW_NUMBER() OVER (PARTITION BY elderly_id ORDER BY appointment_time DESC) rn_desc,

ROW_NUMBER() OVER (PARTITION BY elderly_id ORDER BY appointment_time ASC) rn_asc

FROM appointments a

)

SELECT e.id, e.full_name,

  MAX(CASE WHEN ar.rn_desc = 1 THEN ar.status END) AS last_appt_status,

  MIN(CASE WHEN ar.rn_asc = 1 AND ar.appointment_time > NOW() THEN ar.appointment_time END) AS next_appt_time

FROM appt_rank ar

JOIN elderly_person e ON e.id = ar.elderly_id

GROUP BY e.id, e.full_name;

8.Find family notification ratio per elderly (family_notify communications / total comms) last 30 days

Query:

SELECT f.elderly_id, e.full_name,

  ROUND(100 * SUM(c.comm_type = 'family_notify') / NULLIF(COUNT(c.id),0),2) AS pct_family_notify

FROM family_elderly_link f

JOIN elderly_person e ON e.id = f.elderly_id

LEFT JOIN communications c ON c.elderly_id = f.elderly_id AND c.created_at >= DATE_SUB(NOW(), INTERVAL 30 DAY)

GROUP BY f.elderly_id, e.full_name

ORDER BY pct_family_notify DESC;

9.Find elders who have both a prescription and at least one appointment (join across tables)

Query:

SELECT DISTINCT p.elderly_id, e.full_name,

```
    COUNT(DISTINCT p.id) AS prescriptions_count,
    COUNT(DISTINCT a.id) AS appointments_count
FROM elderly_person e
LEFT JOIN prescriptions p ON p.elderly_id = e.id
LEFT JOIN appointments a ON a.elderly_id = e.id
GROUP BY p.elderly_id, e.full_name
HAVING prescriptions_count > 0 OR appointments_count > 0
ORDER BY prescriptions_count DESC, appointments_count DESC;
```

10.Find emergency alerts and urgent communications mapped (shows recipients and how many urgent comms)

Query:

```
SELECT al.id AS alert_id, al.elderly_id, e.full_name,
    SUM(c.comm_type = 'urgent') AS urgent_comms,
    GROUP_CONCAT(DISTINCT c.to_user_id ORDER BY c.to_user_id) AS recipients
FROM alerts al
LEFT JOIN elderly_person e ON e.id = al.elderly_id
LEFT JOIN communications c ON c.elderly_id = al.elderly_id AND c.created_at >=
al.created_at
WHERE al.type = 'emergency'
GROUP BY al.id, al.elderly_id, e.full_name
ORDER BY urgent_comms DESC, al.created_at DESC;
```

Learning Outcomes of the Project:

1. Technologies and Skills

Python: Through this project, we learned how to use Python to build the backend of a web application. We gained experience working with Flask routes, handling user input, connecting to the database, and sending data to the frontend. We also learned how to use Jinja2 templates to display dynamic information on the website.

MySQL: We learned how to design and manage a relational database using MySQL. This included creating tables, writing SQL queries, and using foreign keys to maintain data relationships. We also learned how to connect MySQL with Python and retrieve or update data safely and efficiently.

HTML & CSS: From the frontend part, we learned how to create clean and organized web pages using HTML and CSS. We used Bootstrap to make the pages responsive and user-friendly. We also improved my skills in designing layouts, styling elements, and presenting data in a clear and visually appealing way.

2. Features and Explanations

1. User Role Management

The system supports multiple user types—Admin, Elderly Person, Caregiver, Doctor, and Family Member. Each role has specific permissions and access levels to ensure secure and organized operations.

2. Real-Time Health Monitoring

Vital signs such as heart rate, blood pressure, and other health metrics are recorded through connected devices or manual input. These readings are instantly available to caregivers, doctors, and family members.

3. Emergency Alert System

The elderly person can trigger an emergency alert through a wearable device or dashboard button. The system immediately notifies the caregiver, family, and doctor to ensure quick response during critical situations.

4. Appointment Scheduling and Management

Doctors and elderly persons can schedule appointments. The system displays upcoming visits, reminders, and follow-up schedules to keep medical care on track.

5. Medication and Prescription Management

Doctors can add or update prescriptions, while caregivers and the elderly can view medication schedules. This helps prevent missed doses and improves treatment accuracy.

6. Activity and Task Tracking

Caregivers can update completed care tasks, monitor daily activities, and record health updates, helping maintain a proper care routine.

7. Family Monitoring Dashboard

Family members have a dedicated dashboard to track the elderly person's health, alerts, and appointments. They stay informed and can communicate with the care team when needed.

8. Secure Data Management

All records—health data, alerts, user details, appointments, and prescriptions—are stored securely. Role-based access control ensures that sensitive information is only visible to authorized users.

9. Communication and Notifications

Real-time notifications are sent during emergencies, appointment updates, and status changes. This ensures that all stakeholders remain connected and respond promptly.

10. System Administration Tools

Admins can add new users, assign roles, manage caregivers and doctors, review system activity, and maintain overall data accuracy. They ensure smooth operation and system health.

Practical Applications of This Project

1. **Real-Time Health Monitoring:**
   The system continuously tracks vital signs, health readings, and emergency alerts, helping caregivers and doctors make quick, informed decisions.

2. **Improved Elderly Safety:**
   Emergency alert triggers ensure rapid response during falls, sudden health changes, or other urgent situations.
3. **Centralized Health Management:**
   All medical records, prescriptions, appointments, and care schedules are stored in one place, reducing manual work and errors.

4. **Enhanced Communication:**
   Family members, caregivers, and doctors can stay connected through shared updates, alert notifications, and activity logs.
5. **Streamlined Care Coordination:**
   Admins can assign roles, manage users, and ensure smooth coordination between doctors, caregivers, and family members.
6. **Remote Support and Monitoring:**
   Family members and doctors can check the elderly person's status from anywhere, enabling remote healthcare supervision.
7. **Data-Driven Decision Making:**
   Collected data helps identify health trends, generates insights, and supports better medical decisions.

Limitations of This Project

1.Device Dependency:
Real-time monitoring relies on sensors or wearable devices, which may malfunction or require regular maintenance.

2.Internet Requirement:
The system must stay connected to function properly; poor connectivity can delay updates or alerts.

3.User Training Needed:
Elderly individuals may require guidance to use the system comfortably, especially if they are not familiar with technology.

4.Data Privacy Concerns:
Storing sensitive medical information demands strong security measures; breaches could compromise personal data.

5.Limited Emergency Assurance:
While alerts notify caregivers and doctors, they cannot physically respond immediately if no one is nearby.

6.System Maintenance:
Regular updates and monitoring are needed to ensure accuracy, prevent bugs, and maintain reliable performance.

Conclusion

The ElderlyCare system provides a smart, integrated solution for supporting the health and safety of elderly individuals. By connecting doctors, caregivers, family members, and administrators through a centralized platform, it improves coordination and ensures timely care. Although it has limitations related to connectivity, device reliability, and data security, the project significantly enhances real-time monitoring, emergency response, and overall wellbeing for elderly individuals. With continued improvements and proper training, this system can become a dependable tool for modern elderly support and healthcare management.

Progress Report

Summary

The project selected for development is the **Elderly Care & Emergency Alert System**, designed to support multiple user roles including Admin, Elderly, Doctor, Caregiver, and Family Members. After identifying the responsibilities and functionalities of each role, all necessary entities were listed, and their attributes were defined. Key database constraints such as primary keys, foreign keys, and participation constraints were added to ensure structural accuracy. An initial ER Diagram was created to illustrate the relationships among entities, followed by several refinements to validate entity integrity and referential integrity. A complete narrative description of the ER Diagram and its relationships was also prepared.

Once finalized, the ER Diagram was converted into the **relational schema**, clearly mapping all primary and foreign key relationships. Each table was tested for normalization, and all were found to satisfy the requirements of **1NF, 2NF, and 3NF**. After confirming proper normalization, the database was created through SQL commands, tables were generated with all constraints, and individual records were inserted into each table as part of the initial data population process.

Group Progress Report

Reporting Period: 1-11-2025 to 30-11-2025

Team Member	Date & Time	Total Hours	Remarks
Ayesha Siddiki Mimi (24549010102)	02.11.2025 — 07:00PM–11:00PM	4 hours	Initial review of ER diagram, refined entity attributes, updated caregiver–elderly relationships
	06.11.2025 — 08:00PM–12:00AM	4 hours	Developed relational schema draft, strengthened table relationships, checked attribute dependencies
	12.11.2025 — 07:30PM–11:30PM	4 hours	Worked on normalization (1NF–3NF), completed detailed explanations

	18.11.2025 — 08:00PM–12:00AM	4 hours	Inserted sample data, validated keys, resolved inconsistencies
	25.11.2025 — 07:00PM–11:30PM	4 hours 30 mins	Updated documentation, ER diagram design, formatted report sections
	29.11.2025 — 08:30PM–12:30AM	4 hours	Final review of database design, wrote chapter summary
Rukaiya Binte Iqbal (24549010104)	01.11.2025 — 08:00PM–11:30PM	3 hours 30 mins	Wrote project introduction, prepared entity & attribute descriptions
	05.11.2025 — 07:15PM–11:15PM	4 hours	Assisted ER diagram revision, added keys, constraints, and cardinalities
	11.11.2025 — 08:00PM–12:00AM	4 hours	Entered data into tables, validated foreign key relations
	17.11.2025 — 07:00PM–11:00PM	4 hours	Helped finalize normalization explanations and dependency analysis
	22.11.2025 — 06:00PM–11:00PM	5 hours	Reviewed table structure, corrected anomalies, strengthened relationship notes
	27.11.2025 — 08:30PM–12:00AM	3 hours 30 mins	Completed constraint documentation and 3NF validation
	30.11.2025 — 02:00PM–08:00PM	6 hours	Finalized full documentation, prepared presentation slides

Group total working hours	—	**54 hours 30 mins**	—

Individual Progress Report

Member Name: Ayesha Siddiki Mimi (24549010102)

Reporting Period: 01.11.2025 – 30.11.2025

Date	Hours	Activities
02.11.2025	4 hours	Reviewed ER Diagram concepts, refined attributes and mapped elderly–caregiver relationships
06.11.2025	4 hours 15 minutes	Developed relational schema structure, corrected functional dependencies, ensured proper PK–FK mapping
09.11.2025	3 hours 40 minutes	Worked on normalization (1NF → 3NF), documented explanations and examples
12.11.2025	4 hours	Entered data into multiple tables, tested constraints and referential integrity
18.11.2025	5 hours	Group meeting, schema revision, identified anomalies, improved table structure
21.11.2025	3 hours 10 minutes	Updated ER diagram after review, added cardinalities and participation constraints
25.11.2025	4 hours 30 minutes	Formatting project documentation, reorganizing chapters, preparing database design description
29.11.2025	4 hours	Final review of relational schema, completed summary of schema benefits and system flow

Member Name: Rukaiya Binte Iqbal (24549010104)

Reporting Period: 01.11.2025 – 30.11.2025

Date	Hours	Activities
01.11.2025	3 hours 45 minutes	Studied ER model concepts from lecture notes and online tutorials
05.11.2025	4 hours	Revised table designs, assisted in drawing updated ER Diagram with relationships and keys
08.11.2025	4 hours 20 minutes	Entered sample data into tables, checked attribute consistency and domain correctness
13.11.2025	5 hours	Group meeting and documentation work, reviewed design decisions and entity mapping
17.11.2025	4 hours 10 minutes	Worked on normalization, explained partial and transitive dependencies
22.11.2025	5 hours 30 minutes	Wrote introduction and functional descriptions for project chapters
27.11.2025	3 hours 40 minutes	Completed documentation of constraints, ensured all tables satisfy 3NF
30.11.2025	6 hours	Final compilation of project report, helped in preparing presentation slides